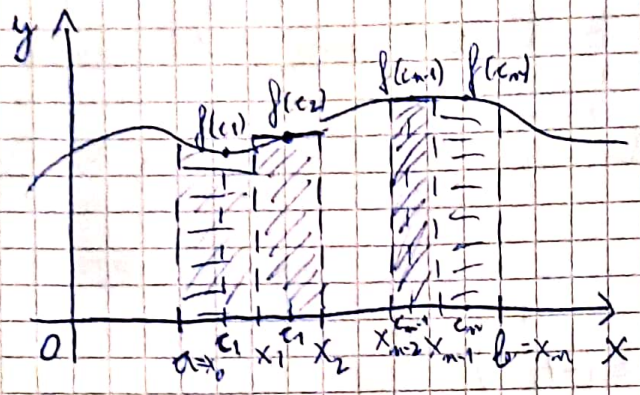
Lecture 7



$\int_a^b f(x)\,dx$ = Area (Signed)

Let $f:[a,b]\to\mathbb{R}$. Take a partition $\Delta: a=x_0, x_1, \dots, x_n=b$ of $[a,b]$

Take intermediate point, $c_k \in [x_{k-1}, x_k]$

Area between x_{k-1} and $x_k \approx$ Area of the rectangle $= f(c_k)\cdot(x_k - x_{k-1})$

$$\text{Area} = \int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1}) = \text{Sum of rectangles}$$

$$\|\Delta\| = \max_{k=\overline{1,n}} (x_k - x_{k-1})$$

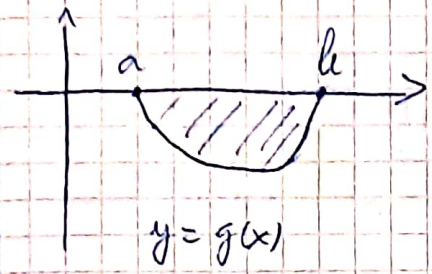
Def: $f:[a,b]\to\mathbb{R}$ is Riemann integrable if for any partition $\Delta = \{x_0, x_1, \dots, x_n\}$ of $[a,b]$ and for any intermediate points $c_k \in [x_{k-1}, x_k]$, $k=\overline{1,n}$ if $\|\Delta\|\to 0$ as $n\to\infty$ we have that

$$\exists \lim_{n\to\infty} \underbrace{\sum_{k=1}^{n} f(c_k)(x_k - x_{k-1})}_{\text{Riemann Sum}} = \int_a^b f(x)\,dx$$

($x_k - x_{k-1} > 0$)

Remark If $x_k - x_{k-1} = \frac{b-a}{n}$, $\forall k=\overline{1,n}$, we get that

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{b-a}{n} f(c_k)$$

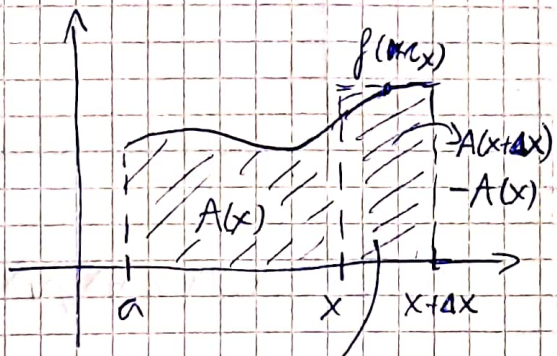


$$\int_a^b g(x)\,dx = -\text{Area}$$

Let $A(x) := \int_a^x f(t)\,dt$

$$A(x+\Delta x) - A(x) \simeq f(c_x)\cdot \Delta x$$

$$A'(x) = \lim_{\Delta x \to 0} \frac{A(x+\Delta x) - A(x)}{\Delta x} = f(x)$$



$$\simeq f(c_x)\cdot \Delta x$$

(Rectangle)

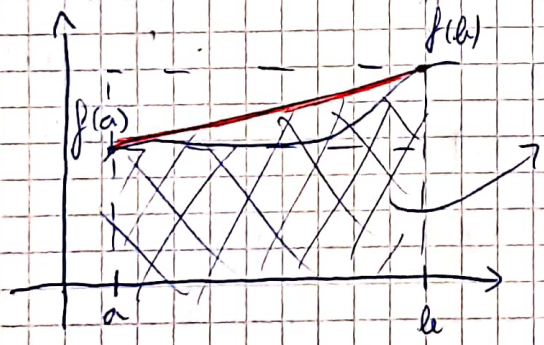
$\Delta x \to 0$, $c_x \to x$

Fundamental Thm of Calculus

Let $F:[a,b]\to\mathbb{R}$ a primitive of f, i.e $F'(x) = f(x)$

Then $\int_a^b f(x)\,dx = F(b) - F(a)$

Trapezium Rule:



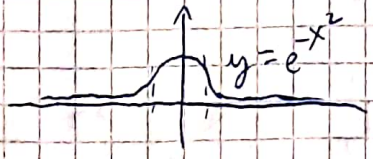
$$\frac{f(a)+f(b)}{2}\cdot(b-a)$$

$$\int_a^b f(x) = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f(x)\,dx \simeq \sum_{k=1}^{n} \frac{f(x_{k-1})+f(x_k)}{2}\,(x_k - x_{k-1})$$

Take $x_k - x_{k-1} = \frac{b-a}{n}$

$$= \frac{(b-a)}{2n}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)\right]$$

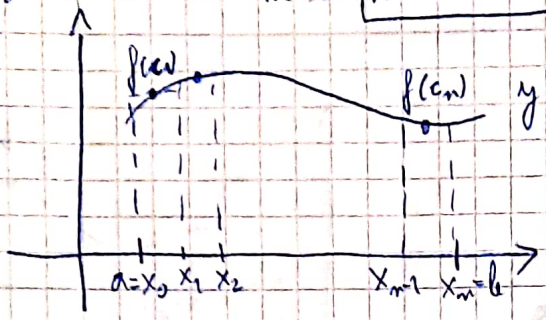
HW!



Seminar 7

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(c_k)(x_k - x_{k-1})$$

↳ Riemann Sum



$c_k \in [x_{k-1}, x_k]$

~~$\sum_{k=1}^{n} f(c_k)(x_k - x_{k-1})$~~

ex: Take $x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1} = \frac{b-a}{n}$

Riemann Sum: $\sum_{k=1}^{n} \frac{b-a}{n} f(c_k)$

ex: $a=0,\ b=1,\ x_k = \frac{k}{n} = c_k$ "] can choose where c_k is on that interval"

$$\int_0^1 f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{n} f\left(\frac{k}{n}\right)$$

(ex 1) a) $\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}\right) = \ln 2$

$$\sum_{k=1}^{n} \frac{1}{n+k} = \sum_{k=1}^{n} \frac{1}{n\left(1+\frac{k}{n}\right)} = \frac{1}{n} \sum_{k=1}^{n} \frac{1}{1+\frac{k}{n}} \xrightarrow{n\to\infty} \int_0^1 \frac{1}{1+x}\,dx$$

$$= \ln(1+x)\Big|_0^1 = \ln 2$$

" $\frac{k}{n}$ will go from 0 (smaller value) to $\frac{n}{n} = 1$ (bigger value) "

c) $\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n} = e^{-1} = \frac{1}{e}$ ("$\ln x_n \to -1 \Rightarrow x_n \to e^{-1}$")

$$\sqrt[n]{\frac{n!}{n^n}} = \sqrt[n]{\frac{1\cdot 2\cdot \dots \cdot n}{n\cdot n\cdot \dots \cdot n}} = x_n$$

$\ln(a\cdot b) = \ln(a) + \ln(b)$

$$\ln x_n = \ln\left(\frac{1}{n}\cdot\frac{2}{n}\cdot\dots\cdot\frac{n}{n}\right)^{\frac{1}{n}} = \frac{1}{n}\ln\left(\frac{1}{n}\cdot\frac{2}{n}\cdot\dots\cdot\frac{n}{n}\right)$$

$$= \frac{1}{n}\sum_{k=1}^{n} \ln\left(\frac{k}{n}\right) \xrightarrow[n\to\infty]{} \int_0^1 \ln(x)\,dx$$

$$\int_0^1 \ln(x)\,dx = \int_0^1 \ln(x)\cdot x'\,dx = \underbrace{x\ln x\Big|_0^1}_{=0} - \underbrace{\int_0^1 x\cdot\frac{1}{x}\,dx}_{=1} = \boxed{-1}$$

$(f\cdot g)' = f'g + fg'$

$f'g = (fg)' - fg' \overset{5}{\Rightarrow} \int_a^b f'(x)g(x)\,dx$

$= f(x)\cdot g(x)\Big|_a^b - \int_a^b f(x)g'(x)\,dx$

$\lim_{x\downarrow 0} x\ln x \overset{0\cdot(-\infty)}{=} \lim_{x\downarrow 0} \frac{\ln x}{\frac{1}{x}} \overset{H}{=} \lim_{x\downarrow 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\downarrow 0}(-x) = 0$

"any x_k I choose, I have to get the same limit to be integrable"

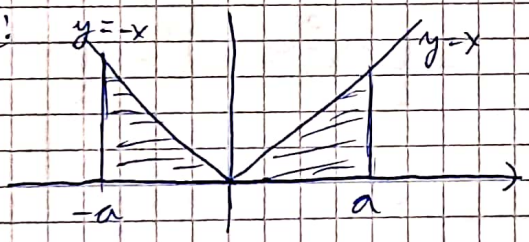
② $f:[0,1]\to\mathbb{R}$, $f(x) = \begin{cases} 1, & x\in\mathbb{Q} \\ 0, & x\in\mathbb{R}\setminus\mathbb{Q} \end{cases}$

a) Take $x_k\in\mathbb{Q}$, $\sum_{k=1}^{n} \underbrace{f(x_k)}_{=1}(x_k - x_{k-1}) = \sum_{k=1}^{n}(x_k - x_{k-1}) = 1$

b) Take $x_k\in\mathbb{R}\setminus\mathbb{Q}$, $\sum_{k=1}^{n} \underbrace{f(x_k)}_{=0}(x_k - x_{k-1}) = 0$

$\nexists \int_0^1 f(x)\,dx$

ex:



"$f(x)=|x|$ not differentiable, but it is integrable"

$\int_{-a}^{a} |x|\,dx = a^2$

Improper Integrals

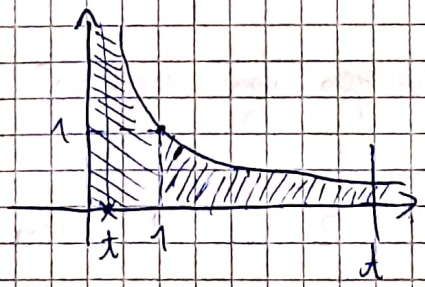
ex: $\int_1^\infty \frac{1}{x^2}\,dx$

$= \lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx$

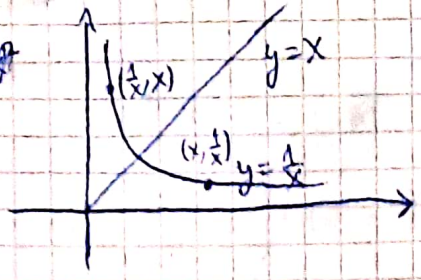
$= \lim_{t\to\infty} \left(-\frac{1}{x}\right)\Big|_1^t$

$= \lim_{t\to\infty} \left(-\frac{1}{t} + 1\right)$

$= 1$



ex: $\int_0^1 \frac{1}{x^2}\,dx = \lim_{t\downarrow 0} \int_t^1 \frac{1}{x^2}\,dx = \lim_{t\downarrow 0} \left(-\frac{1}{x}\right)_t^1 = \lim_{t\downarrow 0}\left(-1 + \frac{1}{t}\right) = \infty$



$y = f(x)$

swap $x \leftrightarrow y$

$x = f(y) \Rightarrow \dots \; y = f^{-1}(x)$

$y = \frac{1}{x^2}$, $x = \frac{1}{y^2} \Rightarrow y^2 = \frac{1}{x} \Rightarrow y = \sqrt{\frac{1}{x}}$

$x \leftrightarrow y$

NOT ~~Symmetric~~ Symm.

(3) a) $\int_1^2 \frac{1}{x(x-2)}\,dx = \frac{1}{2}\int_1^2 \left(\frac{1}{x} - \frac{1}{x-2}\right)dx = -\frac{1}{2}\int_1^2 \frac{1}{x}\,dx + \frac{1}{2}\int_1^2 \frac{1}{x-2}\,dx$

$= -\frac{1}{2}\ln x\Big|_1^2 + \frac{1}{2}\ln|x-2|\Big|_1^2 = -\infty$

$\ln 0 = -\infty$

$[\ln(2-x)]' = \frac{-1}{2-x} = \frac{1}{x-2}$

b) $\int_0^\infty x e^{-x^2}\,dx = -\frac{1}{2}e^{-x^2}\Big|_0^\infty = -\frac{1}{2}(\underbrace{e^{-\infty}}_{=0} - \underbrace{e^0}_{=1}) = \frac{1}{2}$

$(e^{-x^2})' = -2x e^{-x^2}$

$x e^{-x^2} = -\frac{1}{2}(e^{-x^2})'$

ex: $\int_1^\infty x^\alpha\,dx = \frac{x^{\alpha+1}}{\alpha+1}\Big|_1^\infty$

$\alpha \neq -1$

Converges for $\alpha + 1 < 0$; $\alpha < -1$

Comparison Test

If $\frac{f(x)}{x^\alpha} \to l \in \mathbb{R}$ (finite), then $\int_1^\infty f(x)\,dx$ "like" $\int_1^\infty x^\alpha\,dx$

↑ "you can take any other function"

(4) a) $\int_1^\infty \frac{1}{x\sqrt{1+x^2}}\,dx$

$\frac{1}{x\sqrt{1+x^2}} \sim x^{-1}\cdot x^{-1} = x^{-2}$

(comparison test)

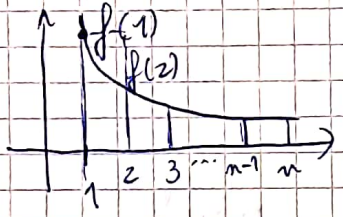
$$\lim_{x\to\infty} \frac{x^2}{x\sqrt{1+x^2}} = 1$$

Serie $\int_1^\infty x^{-2}\,dx$ converge

$\Rightarrow \int_1^\infty \frac{1}{x\sqrt{1+x^2}}\,dx$ is converge

on $\frac{1}{\sin x} \sim \frac{1}{x}$

$f: (0,\infty) \to \mathbb{R}$, decreasing



$\int_1^\infty f(x)\,dx$ "like" $\sum_{n=1}^{\infty} f(n)$ (Integral Test)

$$f(2) \le \int_1^2 f(x)\,dx \le f(1)$$

$$f(3) \le \int_2^3 f(x)\,dx \le f(2)$$

$$\vdots$$

$$f(n) \le \int_{n-1}^n f(x)\,dx \le f(n-1) \quad +$$

$$\sum_{k=2}^{n} f(k) \le \int_1^n f(x)\,dx \le \sum_{k=1}^{n-1} f(k)$$

"If $\sum_{k=2}^{n} f(k)$, $\sum_{k=1}^{n-1} f(k)$ converge $\Rightarrow \int_1^n f(x)\,dx$ converge"

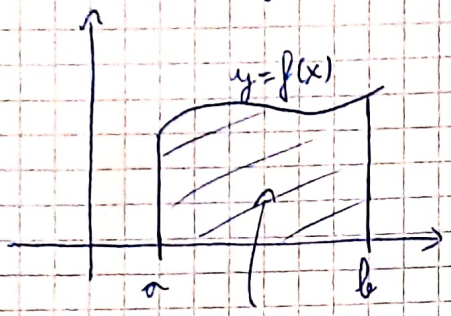
ex: $\int_1^\infty x^\alpha\,dx$ "like" $\sum_{n=1}^{\infty} n^\alpha$

$\sum_{n=1}^{\infty} n^\alpha = \sum_{n=1}^{\infty} \frac{1}{n^r}$; $r = -\alpha$

converges if $\alpha < -1$, $\boxed{r > 1}$

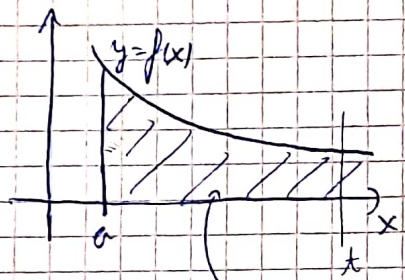
Course 8

Riemann integral

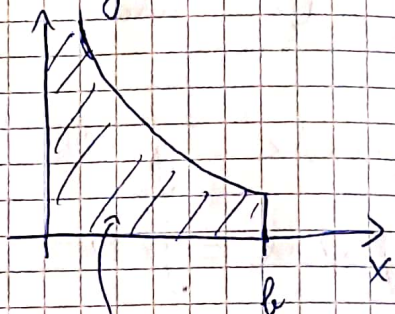


(Signed) Area $= \int_a^b f(x)\,dx$

Improper integral (infinite region)

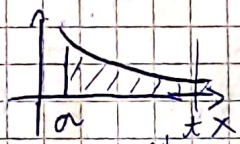


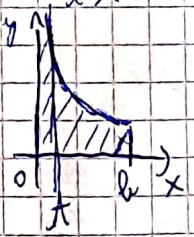
$\int_a^{\infty} f(x)\,dx$



$\int_{a+0}^{b} f(x)\,dx$

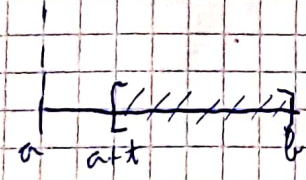
"a+0 - you are getting close to a and at a something happens"





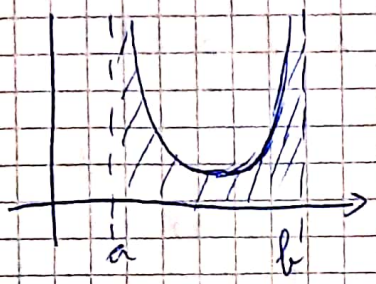
- $\int_a^{\infty} f(x)\,dx := \lim_{t\to\infty} \int_a^t f(x)\,dx$

- $\int_{a+0}^{b} f(x)\,dx = \lim_{t\to a} \int_t^b f(x)\,dx = \lim_{t\searrow 0} \int_{a+t}^{b} f(x)\,dx$



example: a) $\int_1^{\infty} \frac{1}{x^2}\,dx$

$$\int_{a+0}^{\infty} f(x)\,dx$$



$$\int_{a+0}^{b-0} f(x)\,dx$$

example: a) $\int_1^{\infty} \frac{1}{x^2}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx = \lim_{t\to\infty} -\frac{1}{x}\Big|_1^t$

$= \lim_{t\to\infty} \left(-\frac{1}{t}+1\right) = 1$ CONVERGENT

- $\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx \begin{cases} <\infty & \text{CONVERGENT} \\ \pm\infty & \text{DIVERGENT} \end{cases}$
- $\int_{a+0}^{b} f(x)\,dx = \lim_{t\to\infty} \int_t^b f(x)\,dx \begin{cases} <\infty & \text{CONV} \\ \pm\infty & \text{DIV} \end{cases}$

b) $\int_0^1 \frac{1}{x^2}\,dx = \lim_{t\to 0} \int_t^1 \frac{1}{x^2}\,dx = \lim_{t\to 0} \left(-\frac{1}{x}\right)\Big|_t^1$

$(=0+0)$

$= \lim_{t\to 0} \left(-1+\frac{1}{t}\right) = \infty$ DIVERGENT

example: a) $\int_a^{\infty} \frac{1}{x^r}\,dx$, $a>0$. Convergence depends on r. How?

I. $r\neq 1 \Rightarrow \int_a^t \frac{1}{x^r}\,dx = \int_a^t x^{-r}\,dx = \frac{x^{-r+1}}{-r+1}\Big|_a^t = \frac{t^{-r+1}}{-r+1} - \frac{a^{-r+1}}{-r+1}$

$\lim_{t\to\infty} \int_a^t \frac{1}{x^r}\,dx = \lim_{t\to\infty} \left(\frac{t^{-r+1}}{-r+1} - \frac{a^{-r+1}}{-r+1}\right)$

→ CONV. if $-r+1<0 \Rightarrow \boxed{r>1}$

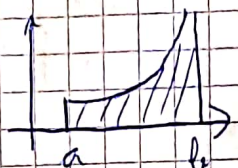
→ Div. if $-r+1\geq 0 \Rightarrow \boxed{r\leq 1}$

II. $p=1 \Rightarrow \int_a^\infty \frac{1}{x}\,dx = \lim_{t\to\infty} \int_a^t \frac{1}{x}\,dx = \lim_{t\to\infty} (\ln t - \ln a)$

$= \infty$

CONCLUSION: $\int_a^\infty \frac{1}{x^p}\,dx \begin{cases} \text{CONV if } p>1 \\ \text{DIV if } p\le 1 \end{cases}$

b) $\int_a^{b-0} \frac{1}{(b-x)^p}\,dx$



$= \lim_{t\to 0} \int_a^{b-t} \dots =$

I. $p\neq 1$: $\int_a^b \frac{1}{(b-x)^p}\,dx = \int_a^b (b-x)^{-p}\,dx = \lim_{t\to 0} \frac{-(b-x)^{-p+1}}{-p+1}\Big|_a^{b-t}$

$= -\lim_{t\to 0} \left(\frac{t^{-p+1}}{-p+1} - \frac{(b-a)^{-p+1}}{-p+1} \right)$

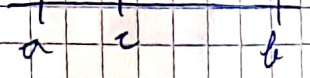
CONV. if $-p+1>0 \Rightarrow \boxed{p<1}$ DIV if $-p+1<0 \Rightarrow \boxed{p>1}$

II $p=1$: HW.

Prop (Comparison test)

Let $f, g : [a,b) \to [0,\infty)$, $b \in (a,\infty]$

If $f(x) \le g(x)$, $\forall x \in (c,b)$ then



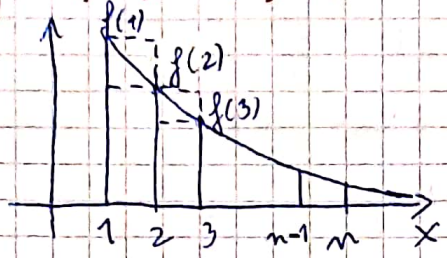
a) $\int_a^{b-0} g(x)\,dx$ CONV $\Rightarrow \int_a^{b-0} f(x)\,dx$ CONV

b) $\int_a^{b-0} f(x)\,dx$ DIV $\Rightarrow \int_a^{b-0} g(x)\,dx$ DIV

Prop $\lim_{\substack{x\to b \\ b\in[0,\infty]}} \frac{f(x)}{g(x)} \in (0,\infty)$, then $\int_a^{b-0} f(x)\,dx$ "like" $\int_a^{b-0} g(x)\,dx$

Integral Test

Let $f:[1,\infty)\to(0,\infty)$ decreasing



$$\int_1^\infty f(x)\,dx = \lim_{n\to\infty}\int_1^n f(x)\,dx$$

$$\int_1^n f(x)\,dx = \int_1^2 f(x)\,dx + \int_2^3 f(x)\,dx + \dots + \int_{n-1}^n f(x)\,dx$$

$$f(2) \le \int_1^2 f(x)\,dx \le f(1)$$

$$f(3) \le \int_2^3 f(x)\,dx \le f(2)$$

$$\vdots$$

$$f(n) \le \int_{n-1}^n f(x)\,dx \le f(n-1) \quad +$$

$$f(2)+f(3)+\dots+f(n) \le \int_1^n f(x)\,dx \le f(1)+f(2)+\dots+f(n-1)$$

$$\sum_{k=2}^{n} f(k) \le \int_1^n f(x)\,dx \le \sum_{k=1}^{n-1} f(k)$$

let $n\to\infty$: $\displaystyle\sum_{k=2}^{\infty} f(k) \le \int_1^\infty f(x)\,dx \le \sum_{k=1}^{\infty} f(k)$

$\int_1^\infty f(x)\,dx$ "like" $\sum_{n=1}^{\infty} f(n)$

ex: a) $\sum_{n=1}^{\infty} \frac{1}{n^r}$ "like" $\int_1^\infty \frac{1}{x^r}\,dx$ → conv if $r>1$, div if $r\le 1$

b) In Seminar

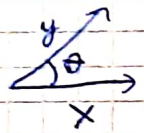
$$\sum_{n\ge 2} \frac{\ln n}{n^2} \qquad \int_1^\infty \frac{\ln x}{x^2}\,dx$$

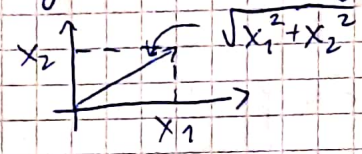
Chapter 3 (?)

- The space $\mathbb{R}^n$, $n \in \mathbb{N}$
- $x \in \mathbb{R}^n$, $x = (x_1, x_2, \dots, x_n) = \sum_{k=1}^{n} x_k e_k$
- $x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $\alpha x = (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$
  $\alpha \in \mathbb{R}$
- $e_1 = (1, 0, \dots, 0)$; $e_2 = (0, 1, \dots, 0)$, $\dots$, $e_n = (0, 0, \dots, 1)$
- Scalar product

(Dot Product) $x \cdot y \overset{\text{def}}{=} x_1 y_1 + x_2 y_2 + x_n y_n$

Ex: $n \in \{2, 3\}$, $x \cdot y = \cos(\theta) \cdot \text{Length}(x) \cdot \text{Length}(y)$





$$\text{Length}(x) = \underbrace{\sqrt{x_1^2 + x_2^2}}_{(2D)} \text{ or } \underbrace{\sqrt{x_1^2 + x_2^2 + x_3^2}}_{(3D)}$$

$x \perp y \iff x \cdot y = 0$

Properties: 1) $x \cdot y = y \cdot x$

2) $(\alpha x + \beta y) \cdot z = \alpha x z + \beta y z$

3) $x \cdot x \geq 0$

Def: $\langle \cdot, \cdot \rangle$ (two arguments) is a scalar product if it satisfies

1) $\langle x, y \rangle = \langle y, x \rangle$

2) $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$

3) $\langle x, x \rangle \geq 0$

Def: $\|x\| = \sqrt{\langle x, x \rangle}$ ↳ Norm of x

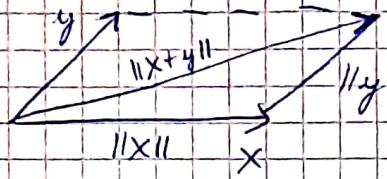
(Dot Product) $\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$ Euclidean Norm

Properties: (1) $\|x\| = 0 \Rightarrow x = 0_n$

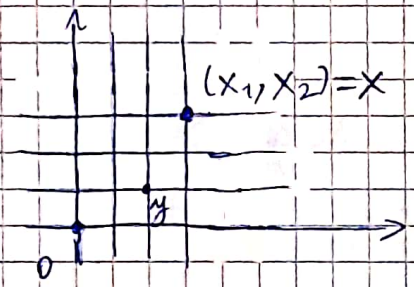
2) $\|\alpha x\| = |\alpha| \|x\|$

3) $\|x+y\| \leq \|x\| + \|y\|$ (Triangle inequality)

$\|x\|$ = length of a vector ("in a weird way")



If $\|\cdot\|$ satisfies the properties above, $\|\cdot\|$ is called a norm



$$\|x\| = |x_1| + |x_2|$$

↑ Manhattan Norm

$$x \cdot y = \cos(\theta)\, \|x\|\, \|y\|$$

Cauchy-Schwarz inequality

$$\langle x, y \rangle \leq \|x\|\, \|y\|$$

Ex: $n \in \{2, 3\}$, take the dot product

$$x \cdot y = \underbrace{\cos(\theta)}_{\leq 1}\, \|x\|\, \|y\| \leq \|x\|\, \|y\|$$

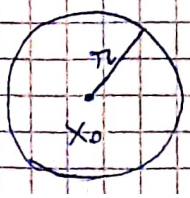
Def: Distance $d(x, y) :\overset{\text{def}}{=} \|x - y\|$

Def: Ball $B(x_0, r) := \{x \in \mathbb{R}^n \mid \|x - x_0\| < r\}$

↳ open ball

$$\overline{B}(x_0, r) := \{x \in \mathbb{R}^n \mid \|x - x_0\| \leq r\}$$

↳ closed ball



Seminar 8

$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$

$y = (y_1, y_2, \dots, y_n)$

$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$

$$x \cdot y = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

↳ Dot Product

$\langle x, y \rangle$ = Scalar Product of x and y

Scalar product satisfies

1) $\langle x, y \rangle = \langle y, x \rangle$

2) $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$

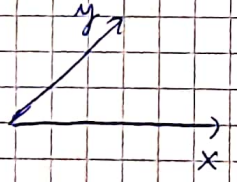
3) $\langle x, x \rangle \geq 0$

Norm: $\|x\| = \sqrt{\langle x, x \rangle}$

$\|x^2\| = \langle x, x \rangle$

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

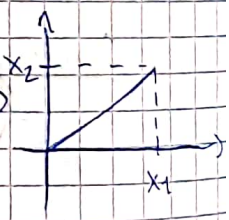
$n = 2$

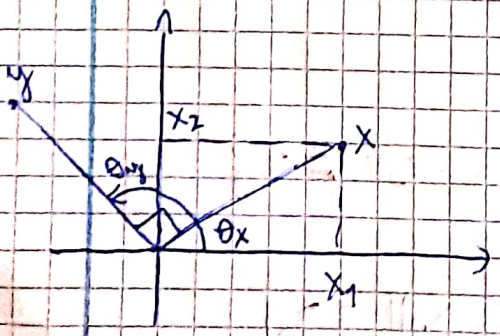


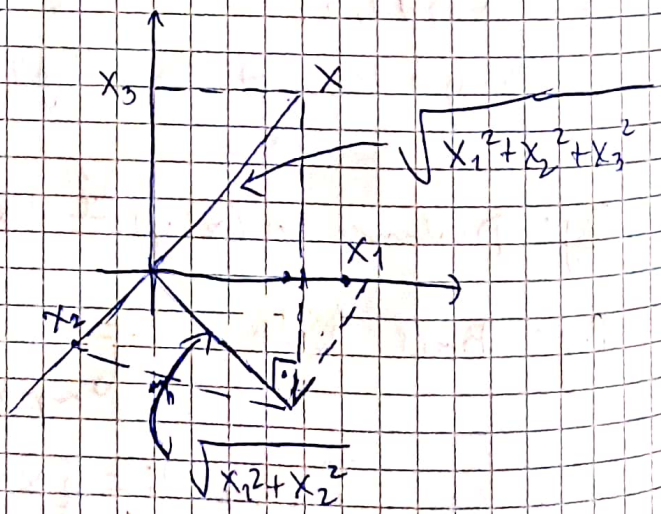
$x \cdot y = x_1 y_1 + x_2 y_2$

$\|x\| = \sqrt{x_1^2 + x_2^2}$

$\|y\| = \sqrt{y_1^2 + y_2^2}$





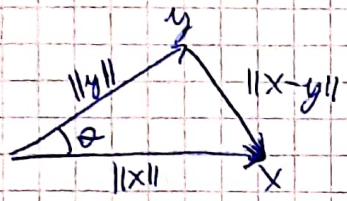


$m_x = \tan(\theta_x) = \frac{x_2}{x_1}$

$m_y = \tan(\theta_y) = \frac{y_2}{y_1}$

$x \perp y \iff m_x \cdot m_y = -1 \iff \frac{x_2 y_2}{x_1 y_1} = -1 \iff x_2 y_2 = -x_1 y_2$

$\iff x_1 y_1 + x_2 y_2 = 0 \iff x \cdot y = 0$



Cosine Rule: $a^2 = b^2 + c^2 - 2bc\cos A$

$\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos(\theta)$ (1)

$\|x-y\|^2 = (x-y)\cdot(x-y) \; (= \langle x-y, x-y\rangle)$

$= x\cdot x - x\cdot y - y\cdot x + y\cdot y$

$= \|x\|^2 - 2x\cdot y + \|y\|^2$ (2)

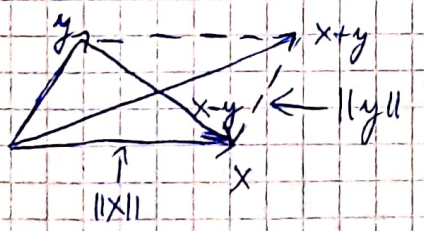
(1),(2) $\Rightarrow$ $\boxed{x\cdot y = \|x\|\|y\|\cos\theta}$

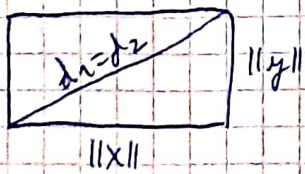
$x \perp y$, $x\cdot y = \|x\|\|y\|\underbrace{\cos(\frac{\pi}{2})}_{=0}$

Unit vectors: $\|x\| = \|y\| = 1$ then $x\cdot y = \cos\theta$

① a) $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$

$d_1^2 + d_2^2 = \sum \text{side}^2$





$d_1^2 + d_2^2 = 2(\|x\|^2 + \|y\|^2)$

- $\|x+y\|^2 = \langle x+y, x+y\rangle = \langle x, x+y\rangle + \langle y, x+y\rangle$

$= \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle$

$= \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$ (*)

• $\|x-y\| = \langle x-y, x-y\rangle$

$= \langle x, x\rangle - \langle x, y\rangle - \langle y, x\rangle + \langle y, y\rangle$

$= \|x\|^2 - 2\langle x, y\rangle + \|y\|^2 \quad (**)$

$(*)+(**)$

$\Rightarrow \|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$

b) $\|x+y\|^2 - \|x-y\|^2 \overset{(*)-(**)}{=\!=} 4\langle x, y\rangle$

$\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$

$x \cdot y = \underbrace{\cos\theta}_{\in[-1,1]} \|x\| \cdot \|y\|$

$-\|x\|\cdot\|y\| \le x \cdot y \le \|x\|\cdot\|y\|$

$n = 2, 3$

$|x \cdot y| \le \|x\| \cdot \|y\|$

$|x_1 y_1 + x_2 y_2| \le \sqrt{x_1^2 + x_2^2} \cdot \sqrt{y_1^2 + y_2^2}$

$(x_1 y_1 + x_2 y_2)^2 \le (x_1^2 + x_2^2)(y_1^2 + y_2^2)$

↳ Cauchy - Schwarz

$(x_1 y_1 + \dots + x_n y_n)^2 \le (x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2)$

$(x \cdot y)^2 \le \|x\|^2 \|y\|^2$

3) Cauchy - Schwarz

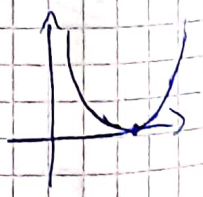
$|\langle x, y\rangle| \le \|x\| \|y\|$

Proof: (beginning)

$f(t) = \langle tx + y, tx + y\rangle = \|tx + y\|^2 \ge 0$

$f(t) = t^2 \langle x, x\rangle + 2\langle x, y\rangle t + \langle y, y\rangle$

$f(t) = \underbrace{t^2 \|x\|^2}_{a \ge 0} + \underbrace{2\langle x, y\rangle}_{b} t + \underbrace{\|y\|^2}_{c} \ge 0$



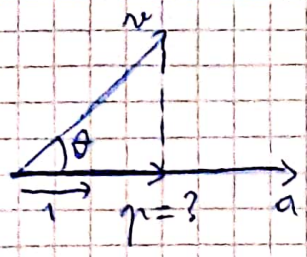
$\Delta \le 0$

$\Delta = 4\langle x, y\rangle^2 - 4\|x\|^2 \|y\|^2 \le 0$

$\langle x, y\rangle^2 \le \|x\|^2 \|y\|^2$

$-\|x\|\|y\| \le \langle x, y\rangle \le \|x\|\|y\|$

$|\langle x, y\rangle| \le \|x\| \|y\|$



$\|p\| = ?$

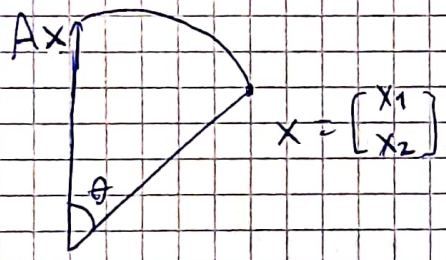
Unit vector: $\frac{1}{\|a\|} \cdot a$

$$p = \frac{1}{\|a\|} \cdot a \cdot \|p\| = \frac{\|p\|}{\|a\|} \cdot a \quad (*)$$

$$\cos\theta = \frac{\|p\|}{\|v\|} = \frac{a \cdot v}{\|a\| \cdot \|v\|} \Rightarrow \|p\| = \frac{a \cdot v}{\|a\|}$$

$$a \cdot v = \cos\theta \, \|a\| \, \|v\| \Rightarrow \cos\theta = \frac{a \cdot v}{\|a\| \|v\|}$$

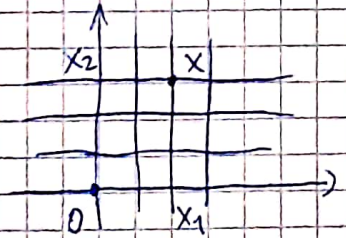
$$(*) \Rightarrow p = \frac{a \cdot v}{\|a\|^2} \cdot a$$



$A^{-1}(Ax) = x$

(6) $\|x\|_p := (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{\frac{1}{p}}$

$p=1$: $\|x\|_1 = |x_1| + |x_2| + \dots + |x_n|$



$|x_1| + |x_2|$

2R 3U

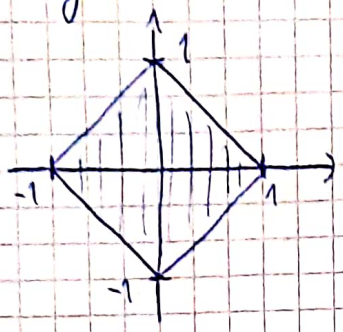
RRUUU ------

$\binom{5}{2} = \binom{5}{3}$ = nr of paths between 0 and x

$B(0,1) = \{x \in \mathbb{R}^2 / \|x\|_p < 1\}$

$p=1$: $|x|_1 = |x_1| + |x_2| < 1$

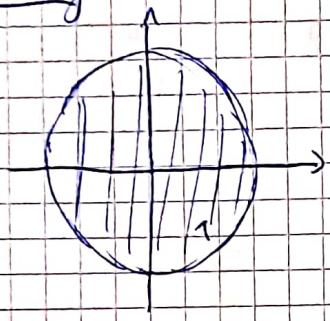
Boundary: $|x_1| + |x_2| = 1$



$p=2$, $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$

$x_1^2 + x_2^2 < 1$

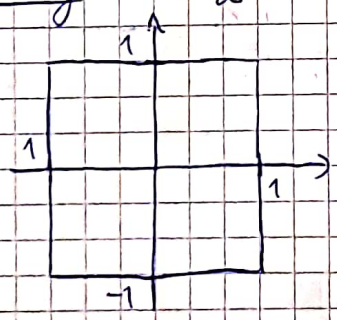
Boundary: $x_1^2 + x_2^2 = 1$



$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$

$n=2$: $\|x\|_\infty = \max\{|x_1|, |x_2|\} < 1$

Boundary: $\|x\|_\infty = \max\{|x_1|, |x_2|\} = 1$



- generate points randomly and see if it's in the plot

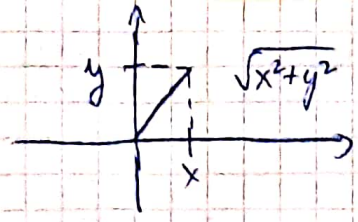
Seminar 9

(1) $c=0$, $L_0 = \{(x,y) \in \mathbb{R}^2 / \sqrt{x^2+y^2} = 0\} = \{(0,0)\}$

$x^2 + y^2 = 0 \Rightarrow x = y = 0$

$c=1$, $L_1 = \{(x,y) \in \mathbb{R}^2 / \sqrt{x^2+y^2} = 1\} = \mathcal{C}(0,1)$

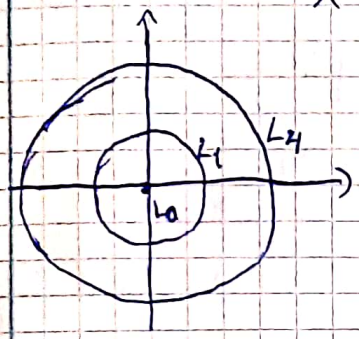
↳ circle of center (0,0) and radius 1



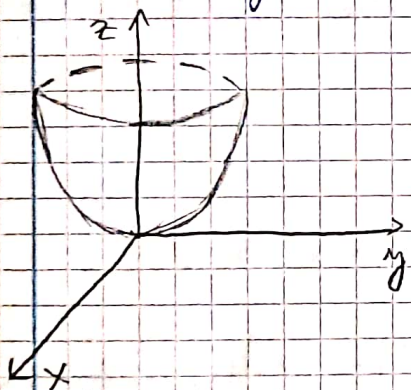
$c=4$, $L_4 = \{(x,y) \in \mathbb{R}^2 \mid \sqrt{x^2+y^2} = 4\} = \mathcal{C}(0,4)$

In general, the equation of a circle is $\sqrt{x^2+y^2} = R$, $x^2+y^2 = 16$

$x^2+y^2 = R^2$



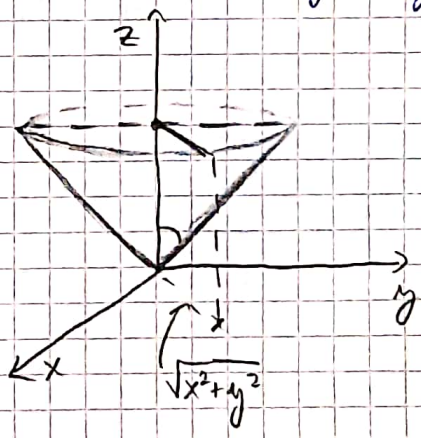
$z = x^2+y^2$



$z = f(x,y) = \sqrt{x^2+y^2}$

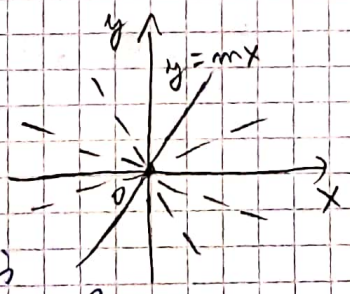
zy-plane: $x=0$

$z = \sqrt{y^2} = |y|$



(2) 1d: $\lim_{x \to 0} f(x)$

2d: $\lim_{(x,y) \to (0,0)} f(x,y)$



$\lim_{(x,mx) \to (0,0)} f(x,y)$ → does this depend on m?

If the limit exists, then we should obtain the same value for all directions

a) Take $y = mx$, check if $\lim_{(x,mx) \to (0,0)} f(x,y)$ depends on m

If depends on $m \Rightarrow \nexists \lim$, else $\exists \lim$

$f(x,y) = \frac{x+y}{x^2+y^2} = \frac{x+mx}{x^2+m^2x^2} = \frac{x(m+1)}{x^2(m^2+1)} = \frac{m+1}{m^2+1}\cdot\left(\frac{1}{x}\right) =$ doesn't exist

$\downarrow \pm\infty$

b) $f(x,y) = \frac{x^2-y^2}{x^2+y^2} = \frac{x^2-m^2x^2}{x^2+m^2x^2} = \frac{x^2(1-m^2)}{x^2(1+m^2)} = \frac{1-m^2}{1+m^2}$ ↑

Take $y = mx$

depends on m ⇓ the limit doesn't exist!

For example, $m=1$ $(y=x)$, $\lim_{(x,x)\to(0,0)} f(x,y) = 0$

$m=2$ $(y=2x)$, $\lim_{(x,2x)\to(0,0)} f(x,y) = \frac{3}{5}$

c) $f(x,y) = \frac{x^3+y^3}{x^2+y^2} = \frac{x^3(1+m^3)}{x^2(1+m^2)} = \frac{1+m^3}{1+m^2}\cdot x \to 0$ as $x\to 0$

Take $y = mx$

This doesn't depend on m

The limit could exist $(=0)$

Show that $\lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} = 0$

Goal: $\left|\frac{x^3+y^3}{x^2+y^2}\right| \le$ something that goes to 0

$$\left|\frac{x^3+y^3}{x^2+y^2}\right| = \left|\frac{(x+y)(x^2-xy+y^2)}{x^2+y^2}\right| = |x+y|\underbrace{\left|\frac{x^2+y^2}{x^2+y^2} - \frac{xy}{x^2+y^2}\right|}_{\text{Bounded?}}$$

$$\le c\underbrace{|x+y|}_{\to 0}$$

Bound $\left|\frac{xy}{x^2+y^2}\right|$, Recall AM–GM: $x^2+y^2 \ge 2|xy|$, $\forall x,y\in\mathbb{R}$

$1 \ge 2\left|\frac{xy}{x^2+y^2}\right|$, hence $\left|\frac{xy}{x^2+y^2}\right| \le \frac{1}{2}$

Hence $\left|\frac{x^3+y^3}{x^2+y^2}\right| \le \left(1+\frac{1}{2}\right)|x+y| \Rightarrow \lim_{(x,y)\to(0,0)} \left|\frac{x^3+y^3}{x^2+y^2}\right| = 0$

d) $\frac{\sin x - \sin y}{x-y}$

$$\lim_{(x,y)\to(0,0)} \frac{\sin x - \sin y}{x-y} \overset{\frac{0}{0}}{=} \lim_{(x,y)\to(0,0)} \frac{2\left[\sin\frac{x-y}{2}\right]\cos\frac{x+y}{2}}{x-y}$$

$$= \lim_{(x,y)\to(0,0)} \frac{2\left[\sin\frac{x-y}{2}\right]\cos\frac{x+y}{2}}{\left[\frac{x-y}{2}\right]\cdot 2} \qquad \lim_{t\to 0}\frac{\sin t}{t} = 1$$

$$= \lim_{(x,y)\to(0,0)} \frac{2\cos\frac{x+y}{2}}{2} = \cos 0 = 1$$

(3) check if $f(0,0) = \lim_{(x,y)\to(0,0)} f(x,y)$ (continuity)

f is continuous at $(x,y)\neq(0,0)$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{(x,mx)\to(0,0)} \frac{mx^2}{x^2+m^2x^2} = \lim_{(x,mx)\to(0,0)} \frac{m}{m^2+1}$$

Take $y = mx$

Depends on m!

⇓

The limit doesn't exist

⇓

The function is not continuous at (0,0)

Recall: $\frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$

$\frac{\partial f}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$

$f(x,y) = \frac{xy}{x^2+y^2}$, $\frac{\partial f}{\partial x}(0,0) = \lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x}$

$f(0,y) = f(x,0) = \frac{x\cdot 0}{x^2+0} = 0$, $(x,y)\neq(0,0)$ $= 0$

$\frac{\partial f}{\partial y}(0,0) = \lim_{y\to 0} \frac{f(0,y)-f(0,0)}{y} = 0$

Hence f has partial derivatives everywhere

1st: Differentiability ⇒ Continuity

Discontinuous ⇒ Not differentiable

nd: Discontinuous ⇏ Not partially differentiable

Differentiable = ?

(4) $\frac{\partial f}{\partial x}$ → consider f as a function of x only (fix the other variables)

a) $f(x,y) =$

$$\frac{\partial f}{\partial x} = \frac{1}{2\sqrt{x^2+y^2}} \cdot 2x \qquad (\sqrt{u})' = \frac{1}{2\sqrt{u}} \cdot u'$$

$$= \frac{x}{\sqrt{x^2+y^2}}$$

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2+y^2}}$$

b) $f(x,y) = \ln \sqrt{x^2+y^2}$ $\qquad (\ln u)' = \frac{u'}{u}$

$$\frac{\partial f}{\partial x} = \underbrace{\frac{1}{\sqrt{x^2+y^2}}}_{\frac{1}{u}} \cdot \underbrace{\frac{2x}{2\sqrt{x^2+y^2}}}_{u'} = \frac{x}{x^2+y^2} \qquad u = \sqrt{x^2+y^2}$$

$$\frac{\partial f}{\partial y} = \frac{y}{x^2+y^2}$$

c) $f(x,y) = \cos x \cos y - \sin x \sin y = \cos(x+y)$

$$\frac{\partial f}{\partial x} = -\sin x \cos y - \cos x \sin y = -\sin(x+y)$$

$$\frac{\partial f}{\partial y} = -\cos x \sin y - \sin x \cos y = -\sin(x+y)$$

d) $f(x,y,z) = x^2yz + ye^z$

$$\frac{\partial f}{\partial x} = (x^2)' yz + 0 = 2xyz$$

$$\frac{\partial f}{\partial y} = x^2 z + e^z$$

$$\frac{\partial f}{\partial z} = x^2 y + y e^z$$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$ (for ex 5)

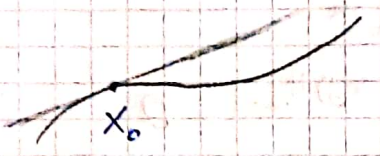
ex 6: Hint: Chain Rule

Let $r = x^2+y^2$, $g(x,y) = f(r)$

Chain Rule: $\frac{\partial g}{\partial x} = \frac{\partial g}{\partial r} \cdot \frac{\partial r}{\partial x}$

Lecture 10 – DIFFERENTIABILITY in $\mathbb{R}^n$

1D:



$$\lim_{x \to x_0} \frac{f(x)-f(x_0)}{x-x_0} = f'(x_0)$$

$$f(x) \approx f(x_0) + \boxed{f'(x_0)(x-x_0)}$$

linear function

$$f(x) = \underbrace{f(x_0) + f'(x_0)(x-x_0)}_{\text{linear approximation}} + R(x-x_0)$$

$$\lim_{x \to x_0} \frac{f(x)-f(x_0)-f'(x_0)(x-x_0)}{x-x_0} = 0$$

with $\lim_{x \to x_0} \frac{R(x-x_0)}{x-x_0} = 0$

$h = x - x_0$, $f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h)-f(x_0)}{h}$

nD: • linear function $\ell(x) = a \cdot x = a_1 x_1 + \dots + a_n x_n$ (*)

$a \in \mathbb{R}^n$ – constant, $x \in \mathbb{R}^n$, $\ell(x) \in \mathbb{R}$

• linear function, $A(x) = A \cdot x \in \mathbb{R}^m$ (**)

$A \in \mathbb{R}^{m \times n}$ $(= [\,]_{m \times n} = M_{m,n})$

$x \in \mathbb{R}^n$

$[\,]_{m \times n} \cdot [\,]_{n \times 1}$

(*) **Def:** Let $f: A \subseteq \mathbb{R}^n \to \mathbb{R}$, $x_0 \in A$. We say that f is differentiable at x_0 if

(2) $$\lim_{x \to x_0} \frac{f(x)-f(x_0)-\boxed{Df(x_0)\cdot(x-x_0)}}{\|x-x_0\|} = 0,$$ where $Df(x_0) \in \mathbb{R}^n$

Notation: $Df(x_0)$ = DIFFERENTIAL of f at x_0

Note that: $Df(x_0)\cdot(x-x_0)$ is a linear function in x

Remark: $f(x) = \underbrace{f(x_0) + Df(x_0)(x-x_0)}_{\text{linear approximation}} + R(x-x_0)$, with $\lim_{x \to x_0} \frac{R(x-x_0)}{\|x-x_0\|} = 0$ (1)

"diff. if it has a linear approx."

In general for a vector-valued function $f: \mathbb{R}^n \to \mathbb{R}^m$

(**) **Def:** $f: A \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $x_0 \in \mathbb{R}^n$. We say that f is differentiable at x_0 if

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - \underbrace{Df(x_0)\cdot(x-x_0)}_{\in \mathbb{R}^m}\|_m}{\|x - x_0\|_n} = 0, \text{ where } Df(x_0) \in \mathbb{R}^{m \times n}$$

$$Df(x_0)\cdot(x-x_0) = [\,]_{m\times n} \cdot [\,]_{n\times 1} = [\,]_{m\times 1}$$

Remark: $f(x) = f(x_0) + Df(x_0)(x-x_0) + R(x-x_0)$, with

$$\lim_{x \to x_0} \frac{\|R(x-x_0)\|}{\|x-x_0\|} = 0$$

Thm $f: A \subseteq \mathbb{R}^n \to \mathbb{R}$ differentiable at x_0 then f is continuous at x_0

Proof: $f(x) = f(x_0) + Df(x_0)(x-x_0) + R(x-x_0)$

(1)=(2) $= f(x_0) + \underbrace{Df(x_0)}_{\text{fixed}}\cdot\underbrace{(x-x_0)}_{\to 0} + \underbrace{\frac{R(x-x_0)}{\|x-x_0\|}}_{\to 0}\cdot\underbrace{\|x-x_0\|}_{\to 0}$

Take $\lim_{x \to x_0} f(x) = f(x_0) + 0 + 0 = f(x_0)$

Hence f is cont. at x_0

Example: (a) $f: \mathbb{R}^n \to \mathbb{R}$, $f(x) = c$ = constant, $f(x) - f(x_0) = 0$

$Df(x) = 0_n = (0, 0, \dots, 0)$

(b) $f: \mathbb{R}^n \to \mathbb{R}$, $f(x) = a \cdot x$, $a \in \mathbb{R}^n$ (vector)

$f(x) - f(x_0) = a(x - x_0)$

$Df(x_0) = a \in \mathbb{R}^n$

"We take Df such that the top of the fraction in (2) is 0"

(c) $f: \mathbb{R}^n \to \mathbb{R}^m$, $f(x) = A\cdot x$, $A \in \mathbb{R}^{m\times n}$, $f(x) - f(x_0) = A(x-x_0)$

$Df(x_0) = A \in \mathbb{R}^{m\times n}$

$$\lim_{x\to x_0} \frac{f(x)-f(x_0)-Df(x_0)(x-x_0)}{\|x-x_0\|}=0$$

or in another notation $h=x-x_0$

$$\lim_{h\to 0} \frac{f(x+h)-f(x)-Df(x)\cdot h}{\|h\|}=0$$

Thm Let $f: A\subseteq \mathbb{R}^n \to \mathbb{R}$ differentiable at $x\in A$. Then the partial derivatives $\frac{\partial f}{\partial x_i}$, $i=\overline{1,n}$, exist and

Δ = Delta

∇ = Nabla

$$\boxed{Df(x)=\nabla f(x)=\left(\frac{\partial f}{\partial x_1}(x),\dots,\frac{\partial f}{\partial x_n}(x)\right)}$$

Proof: Take $h=h_i e_i=(0,\dots,0,\overset{\text{index } i}{h_i},0,\dots,0)$, $\|h\|=h_i>0$

$$0=\lim_{h_i\to 0}\frac{f(x+h_i e_i)-f(x)-Df(x)\cdot h_i e_i}{h_i}$$

$$=\lim_{h_i\to 0}\frac{f(x+h_i e_i)-f(x)}{h_i}-Df(x)\cdot e_i$$

$$=\frac{\partial f}{\partial x_i}(x)-Df(x)_i=0 \quad\Rightarrow Df(x)_i=\frac{\partial f}{\partial x_i}(x)$$

$$\Rightarrow \boxed{Df(x)=\nabla f}$$

$Df(x)\in\mathbb{R}^n$, $Df(x)\cdot e_i=Df(x)_i$

Thm $f:\mathbb{R}^n\to\mathbb{R}^m$ differentiable at $x\in\mathbb{R}^n$ $\quad f=(f_1,\dots,f_m)$

$$Df(x)=\begin{bmatrix}\nabla f_1\\ \dots\\ \nabla f_m\end{bmatrix}=\begin{bmatrix}\frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n}\\ \dots & & \dots\\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n}\end{bmatrix}$$

Thm (Calculus Rules) Let $f,g: A\subseteq\mathbb{R}^n\to\mathbb{R}$, then:

a) $D(f+g)(x)=Df(x)+Dg(x)$

or

$$\nabla(f+g)=\nabla f+\nabla g$$

b) $D(cf)(x)=cDf(x)$

or

$$\nabla(cf)=c\nabla f,\quad c=\text{constant}$$

Product Rule c) $D(fg)(x)=gDf(x)+fDg(x)$

or

$$\nabla(fg) = g\nabla f + f\nabla g$$

Product Rule Proof:

$$\nabla(fg) = \left(\frac{\partial(fg)}{\partial x_1}(x), \dots, \frac{\partial(fg)}{\partial x_n}(x)\right)$$

$$\frac{\partial(fg)}{\partial x_i}(x) = \boxed{\frac{\partial f}{\partial x_i}(x)}\, g(x) + f(x)\, \boxed{\frac{\partial g}{\partial x_i}(x)}$$

↑ standard prod. rule

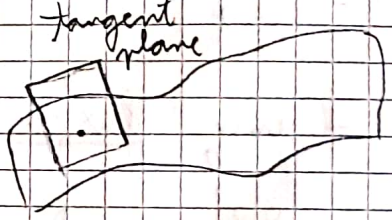
$\frac{\partial f}{\partial x_i}$: component i of ∇f

$\frac{\partial g}{\partial x_i}$: component i of ∇g

$$\nabla(f \cdot g) = g\nabla f + f\nabla g$$

Example: $f: \mathbb{R}^2 \to \mathbb{R}$, $z = f(x,y)$

$$f(x,y) \approx \underbrace{f(x_0,y_0) + Df(x_0,y_0)\cdot(x-x_0, y-y_0)}_{\text{linear approx.}}$$



$$z = z_0 + \nabla f(x_0,y_0)\cdot(x-x_0, y-y_0)$$

$$z = z_0 + \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

↳ equation of the tangent plane at (x_0,y_0)

Line: $ax+by=0$

Plane: $ax+by+cz=0$

Thm (Chain Rule) Let $g: \mathbb{R}^n \to \mathbb{R}^m$, $f: \mathbb{R}^m \to \mathbb{R}^r$

$f\circ g: \mathbb{R}^n \to \mathbb{R}^r$

If g is differentiable at $x \in \mathbb{R}^n$ and f is differentiable at $g(x) \in \mathbb{R}^m$, then $f\circ g$ is differentiable at x and

$$D(f\circ g)(x) = Df(g(x))\cdot Dg(x)$$

$$[\,]_{r\times n} = [\,]_{r\times m}\cdot[\,]_{m\times n}$$

Example: $g: \mathbb{R} \to \mathbb{R}^n$, $g(t) = (\overbrace{g_1(t)}^{x_1}, \dots, \overbrace{g_n(t)}^{x_n})$

$f: \mathbb{R}^n \to \mathbb{R}$, ~~$Df(x) = \nabla f(x) = \left[\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x)\right]$~~

$f\circ g: \mathbb{R} \to \mathbb{R}$, let $h = f\circ g$, $h: \mathbb{R} \to \mathbb{R}$, $h = h(t)$

$h'(t) = Dh(t) = Df(g(t)) \cdot Dg(t)$

$Dg(t) = (g_1'(t), \dots, g_n'(t))$

$Df(x) = \nabla f(x) = \left[\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x)\right]$

$h'(t) = \nabla f(g(t)) \cdot \nabla g(t)$

$= \frac{\partial f}{\partial x_1} \cdot g_1'(t) + \dots + \frac{\partial f}{\partial x_n} g_n'(t)$

$= \frac{\partial f}{\partial x_1} \frac{dx_1}{dt} + \dots + \frac{\partial f}{\partial x_n} \cdot \frac{dx_n}{dt}$

Seminar 10

$f: \mathbb{R}^n \to \mathbb{R}$ is differentiable at x_0

$$\lim_{x \to x_0} \frac{f(x) - f(x_0) - Df(x_0)(x - x_0)}{\|x - x_0\|} = 0$$

$Df(x_0) \in \mathbb{R}^n$, $Df(x_0) = \nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)$

$$\lim_{h \to 0} \frac{f(x+h) - f(x) - Df(x) \cdot h}{\|h\|} = 0, \quad x \in \mathbb{R}^n, \ h \in \mathbb{R}^n$$

① $f(x, y) = xy$

$Df(x_0, y_0) \cdot (x, y) = y_0 x + x_0 y$

$Df(x_0, y_0) = (y_0, x_0)$

$Df(x_1, x_2) = (x_2, x_1)$

Let $x = (x_1, x_2)$, $h = (h_1, h_2)$

$f(x) = x_1 x_2$

$f(x+h) = (x_1 + h_1)(x_2 + h_2)$

$Df(x) \cdot h = (x_2, x_1) \cdot (h_1, h_2) = x_2 h_1 + x_1 h_2$

$$\lim_{(h_1, h_2) \to (0,0)} \frac{|(x_1 + h_1)(x_2 + h_2) - x_1 x_2 - x_2 h_1 - x_1 h_2|}{\sqrt{h_1^2 + h_2^2}}$$

$$\lim_{(h_1, h_2) \to (0,0)} \frac{|h_1 h_2|}{\sqrt{h_1^2 + h_2^2}} \le \lim_{(h_1, h_2) \to (0,0)} \frac{h_1^2 + h_2^2}{2\sqrt{h_1^2 + h_2^2}}$$

$2|ab| \le a^2 + b^2$

$|ab| \le \frac{1}{2}(a^2 + b^2)$

$$\leq \frac{1}{2} \lim_{h \to 0} \sqrt{h_1^2 + h_2^2} = 0$$

$$\boxed{|f(x)| \to 0 \Rightarrow f(x) \to 0}$$

$$\nabla f = (y, x)$$

Directional Derivative, $v \in \mathbb{R}^n$

$$D f_v(x) = \lim_{h \to 0} \frac{f(x + hv) - f(x)}{h}$$

$$D f_v(x) = \nabla f(x) \cdot v$$

- Direction of fastest increase (Steepest ascent) $v = \nabla f$

$u \cdot v = \|u\| \|v\| \cos\theta$

Max: $\cos\theta = 1$, $\theta = 0$

Min: $\cos\theta = -1$, $\theta = \pi$

- Direction of fastest decrease (Steepest descent) $v = -\nabla f$

② $f(x, y) = x^2 + xy$, $(1, 0)$

a) $\nabla f(x, y) = (2x + y, x)$

$\nabla f(1, 0) = (2, 1)$

Steepest descent: $v = -\nabla f(1, 0)$

$v = (-2, -1)$

b) $v = \vec{i} + \vec{j} = (1, 1)$

$D f_v (1, 0) = \nabla f(1, 0) \cdot (1, 1)$

$= (2, 1) \cdot (1, 1) = 2 + 1 = 3 > 0 \Rightarrow$ "the function increases"

$$f(x) \approx \underbrace{f(x_0) + Df(x_0) \cdot (x - x_0)}_{\text{Linear approximation of } f}$$

1D: Tangent line

2D: Tangent plane

$z = f(x, y) = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0)$

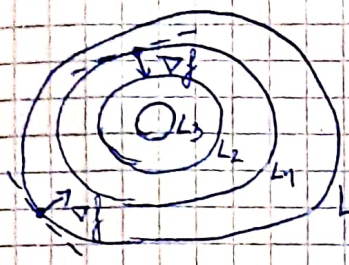
$z = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0)$

Tangent plane at $(1,0,1)$

$z = 1 + 2(x-1) + 1\cdot(y-0)$

$z = 2x + y - 1$

LEVEL SET $Le = \{x \in \mathbb{R}^n / f(x) = c\}$



$\nabla f \perp$ Level Set

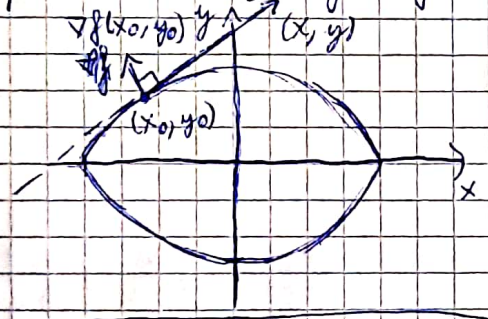
$\nabla f \perp$ Tangent line

$\nabla f \perp$ Tangent plane

③ a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\qquad \nabla f = \left(\frac{2x}{a^2}, \frac{2y}{b^2}\right)$

$f(x,y) = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Ellipse $= L_1 = \{(x,y) / f(x,y) = 1\}$



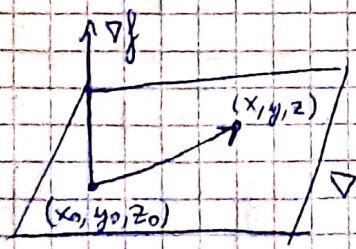
$\nabla f(x_0, y_0) \cdot (x - x_0, y - y_0) = 0$

$\left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}\right) \cdot (x - x_0, y - y_0) = 0$

$\frac{2x_0}{a^2}(x - x_0) + \frac{2y_0}{b^2}(y - y_0) = 0 \;/\cdot \frac{1}{2}$

$\frac{x_0 x}{a^2} - \frac{x_0^2}{a^2} + \frac{y_0 y}{b^2} - \frac{y_0^2}{b^2} = 0$

$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1 \Rightarrow \boxed{\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1}$



$\cdot (x - x_0, y - y_0, z - z_0)$

$\nabla f(x_0, y_0, z_0)$

④ $f: \mathbb{R}^n \to \mathbb{R}$, $f(x) = \|x\|^2 = x_1^2 + \dots + x_n^2$

$x = (x_1, \dots, x_n)$ $\quad \nabla f(x) = (2x_1, \dots, 2x_n) = 2x$

$Df_v(x) = \nabla f(x) \cdot v = 2x \cdot v$

$f(x+hv) = \|x+hv\|^2 = (x+hv)\cdot(x+hv)$

$= x\cdot x + 2hx\cdot v + h^2 v\cdot v$

$= \|x\|^2 + 2hx\cdot v + h^2\|v\|^2$

$Df_v(x) = \lim_{h\to 0} \frac{f(x+hv)-f(x)}{h} = \lim_{h\to 0} \frac{2hx\cdot v + h^2\|v\|^2}{h}$

$= 2x\cdot v + \underbrace{\lim_{h\to 0} h\|v\|^2}_{=0} = 2xv$

⑤ a) $f(x,y) = \ln(x^2+y^2)$, $x = t$, $y = t^2$

$t \to x(t), y(t) \to f(x(t), y(t))$

$\frac{df}{dt} = f'(t) = \frac{2+4t^3}{t^2+t^4}$

$f(t) = \ln(t^2+t^4)$

Chain Rule

$\frac{df}{dt} = \frac{\partial f}{\partial x}\cdot\underbrace{\frac{\partial x}{\partial t}}_{=1} + \frac{\partial f}{\partial y}\cdot\underbrace{\frac{\partial y}{\partial t}}_{2t}$ $\qquad \frac{\partial f}{\partial x} = \frac{2x}{x^2+y^2}$; $\frac{\partial f}{\partial y} = \frac{2y}{x^2+y^2}$

$= \frac{2t}{t^2+t^4}\cdot 1 + \frac{2t^2}{t^2+t^4}\cdot 2t$

BACK PROPAGATION

Lecture 11

Recall that $f: \mathbb{R}^n \to \mathbb{R}$ is differentiable at $x_0 \in \mathbb{R}^n$ if

(*) $\lim_{x\to x_0} \frac{f(x) - f(x_0) - \overbrace{Df(x_0)(x-x_0)}^{\text{linear function}}}{\|x-x_0\|} = 0$; $Df(x_0) \in \mathbb{R}^n$

Thm: If f is differentiable at x_0, then $Df(x_0) = \nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0)\right)$

Thm: If $\frac{\partial f}{\partial x_i}$ (at x_0) exist and are continuous (at x_0), then f is differentiable at x_0

Example: (Seminar 9)
$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$
has partial derivatives at $(0,0)$, but it is not cont at $(0,0)$

Differentiable $\Rightarrow$ continuous
Not continuous $\Rightarrow$ Not differentiable

Def: $f: A \subseteq \mathbb{R}^n \to \mathbb{R}$, a direction $v \in \mathbb{R}^n$. The directional derivative is

$$\boxed{D f_v(x) = \lim_{h \to 0} \frac{f(x+hv) - f(x)}{h}}$$

Thm $f: A \subseteq \mathbb{R}^n \to \mathbb{R}$ differentiable at x, then

$$\boxed{D f_v(x) = \nabla f(x) \cdot v}$$

Proof: f is differentiable at $x \overset{(1)}{\Rightarrow}$

$h = x - x_0 \Rightarrow x = x_0 + h \overset{(*)}{\Rightarrow} \lim_{h \to 0} \frac{f(x_0+h) - f(x_0) - Df(x_0) \cdot h}{\|h\|}$

$\overset{(1)}{\Rightarrow} \lim_{h \to 0} \frac{f(x+hv) - f(x) - \nabla f(x) \cdot hv}{\|hv\|} = 0$

But $\|hv\| = h\|v\|$, hence $\lim_{h \to 0} \frac{f(x+hv) - f(x) - \nabla f(x) \cdot hv}{h\|v\|} = 0$ / $\cdot \|v\|$ (fixed constant)

$\lim_{h \to 0} \frac{f(x+hv) - f(x) - \nabla f(x) \cdot hv}{h} = 0$

$\lim_{h \to 0} \left[\frac{f(x+hv) - f(x)}{h} - \nabla f(x) \cdot v \right] = 0$

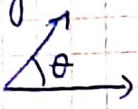
Hence $\lim_{h \to 0} \frac{f(x+hv) - f(x)}{h} = \nabla f(x) \cdot v$

Prop: $f: \mathbb{R}^n \to \mathbb{R}$ differentiable at x, $\nabla f(x) \neq 0$. Then (vector)

- direction of fastest increase (steepest ascent) is $\nabla f(x)$
- direction of fastest decrease (steepest descent) is $-\nabla f(x)$

Proof: $Df_v(x) = \nabla f(x) \cdot v$ and $|\nabla f(x) \cdot v| \leq \|\nabla f(x)\| \cdot \|v\|$ (Cauchy–Schwarz)

$n \in \{2, 3\}$ $\quad \nabla f(x) \cdot v = \|\nabla f(x)\| \, \|v\| \underbrace{\cos(\theta)}_{\in [-1,1]}$



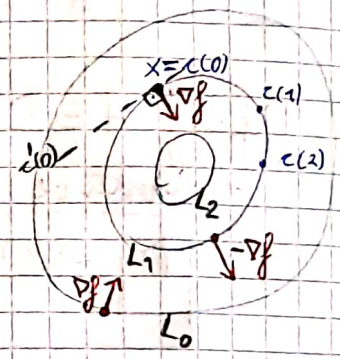
Max for $v = \nabla f$

Min for $v = -\nabla f$

$\cos\theta = 1 \Rightarrow \theta = 0, \; v = \nabla f(x)$

$\cos\theta = -1 \Rightarrow \theta = \pi, \; v = -\nabla f(x)$

Thm ∇f is perpendicular to the level set $L_c = \{x \in \mathbb{R}^n / f(x) = c\}$



Proof

Let $c(t)$ be a path on the level set, $c(0) = x$, $c(t) \in L_c$

$f(c(t)) = \text{constant} \Rightarrow \frac{d}{dt} f(c(t)) = 0$

$t \longmapsto c(t) \in \mathbb{R}^n \longmapsto f(c(t))$

Chain Rule: $D(f \circ g)(x) = Df(g(x)) \cdot Dg(x)$

$\frac{d}{dt} f(c(t)) = \nabla f(c(t)) \cdot c'(t) = 0$

$c'(0)$ = tangent to the level curve at $x = c(0)$

Take $t = 0$: $\nabla f(c(0)) \cdot c'(0) = 0$

$\nabla f(x) \cdot v = 0$,

v - tangent vector

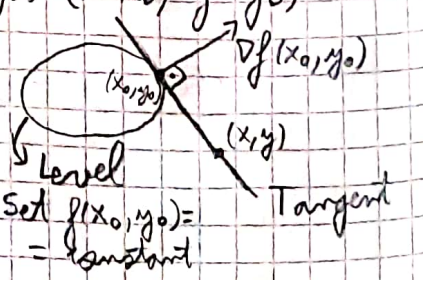
$c(t) = (c_1(t), \ldots, c_n(t)) \in \mathbb{R}^n$

$c'(t) = Dc(t) = (c_1'(t), \ldots, c_n'(t)) \in \mathbb{R}^n$

Let's call the tangent vector $v = c'(0)$

Prop • Tangent line equation: $\nabla f(x_0, y_0) \cdot (x - x_0, y - y_0) = 0$

$$\boxed{\frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) = 0}$$



• Tangent plane equation: $\nabla f(x_0, y_0, z_0)\cdot(x-x_0, y-y_0, z-z_0)$.

Higher order derivatives

Def $\dfrac{\partial^2 f}{\partial x_i \partial x_j} = \dfrac{\partial}{\partial x_i}\left(\dfrac{\partial f}{\partial x_j}\right)$

↳ Mixed Partial derivative

ex: $f(x,y) = xy + (x+2y)^2$

$\dfrac{\partial f}{\partial x} = y + 2(x+2y)$, $\dfrac{\partial^2 f}{\partial x^2} = 2$; $\dfrac{\partial^2 f}{\partial y \partial x} = 5$

$\dfrac{\partial f}{\partial y} = x + 2(x+2y)\cdot 2$; $\dfrac{\partial^2 f}{\partial y^2} = 8$; $\dfrac{\partial^2 f}{\partial x \partial y} = 5$

Thm $f: \mathbb{R}^n \to \mathbb{R}$ of class C^2 (second-order partial derivatives are continuous)

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}, \quad i \neq j$$

Def (HESSIAN matrix) $f: \mathbb{R}^n \to \mathbb{R}$

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_n}\right) = Df(x)$$

$$H(x) = D^2 f(x) = \begin{bmatrix} \nabla \frac{\partial f}{\partial x_1} \\ \vdots \\ \nabla \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

Thm $f(x) = f(x_0) + \underbrace{\nabla f(x_0)\cdot(x-x_0) + \frac{1}{2}(x-x_0)\cdot H(x_0)(x-x_0)}_{n=1:\ f'(x_0)(x-x_0) \quad n=1:\ \frac{f''(x_0)}{2}(x-x_0)^2} + \underbrace{\cdots}_{\text{Remainder}}$

(Taylor Expansion)

Note that: $\underbrace{(x-x_0)}_{\text{vect.}}\cdot\underbrace{H(x_0)}_{\text{matrix}}\underbrace{(x-x_0)}_{\text{vect.}} = (x-x_0)^T H(x_0)(x-x_0)$

$= []_{1\times n}\,\underbrace{[]_{n\times n}\,[]_{n\times 1}}_{[]_{n\times 1}}$

$= []_{1\times n}\cdot[]_{n\times 1} = []_{1\times 1}$

Local extremum point (Max, Min)

Thm If $f:\mathbb{R}^n \to \mathbb{R}$, x_0 - local extremum. Then (Fermat) $\nabla f(x_0) = 0$

- If $\nabla f(x_0) = 0 \Rightarrow f(x) = f(x_0) + \underbrace{(x-x_0)^T H(x_0)(x-x_0)}_{\geq 0 ?\ \leq 0 ?} +$ SMALL REMAINDER

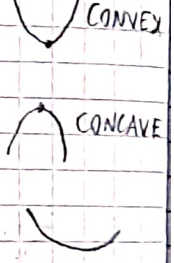
① $(x-x_0)^T H(x_0)(x-x_0) \geq 0 \Rightarrow f(x) \geq f(x_0) \Rightarrow x_0 = MIN$

② $(x-x_0)^T H(x_0)(x-x_0) \leq 0 \Rightarrow f(x) \leq f(x_0) \Rightarrow x_0 = MAX$

Def Matrice $A \in \mathbb{R}^{n\times n}$ is called:

- positive definite if $x^T A x > 0$, $\forall x \in \mathbb{R}^n$ ($\Leftarrow$ Eigenvalues >0)
- negative definite if $x^T A x < 0$, $\forall x \in \mathbb{R}^n$ ($\Leftarrow$ Eigenvalues <0)

Thm $\nabla f(x_0) = 0$ (x_0 is a critical point). Then



① if $H(x_0)$ is positive definite $\Rightarrow x_0 = MIN$

② if $H(x_0)$ is negative definite $\Rightarrow x_0 = MAX$

③ if $H(x_0)$ is indefinite $\Rightarrow x_0 =$ SADLE POINT

Seminar 11

① $f(x,y) = \frac{x}{y} + x\sqrt{y}$, $y>0$

$$\frac{\partial f}{\partial x} = \frac{1}{y} + \sqrt{y} \qquad \frac{\partial f}{\partial y} = -\frac{x}{y^2} + \frac{x}{2\sqrt{y}}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = -\frac{1}{y^2} + \frac{1}{2\sqrt{y}}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = -\frac{1}{y^2} + \frac{1}{2\sqrt{y}}$$

Schwarz Thm:

$\mathbb{R}^2$: $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ | $\mathbb{R}^n$: $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$

2nd order Taylor Polynomial

$\mathbb{R}$: $f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2$

$\mathbb{R}^n$: $f(x) \approx f(x_0) + \nabla f(x_0)\cdot(x-x_0) + \frac{1}{2}(x-x_0)^T H(x_0)(x-x_0)$

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$$\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \frac{\partial f}{\partial x_2}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0)\right)$$

$$H(x_0) = \begin{bmatrix} \nabla\frac{\partial f}{\partial x_1} \\ \nabla\frac{\partial f}{\partial x_2} \\ \dots \\ \nabla\frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x_1 \partial x_1} & \frac{\partial f}{\partial x_2 \partial x_1} & \dots & \frac{\partial f}{\partial x_n \partial x_1} \\ \frac{\partial f}{\partial x_1 \partial x_2} & \frac{\partial f}{\partial x_2 \partial x_2} & \dots & \frac{\partial f}{\partial x_n \partial x_2} \\ \frac{\partial f}{\partial x_1 \partial x_n} & \frac{\partial f}{\partial x_2 \partial x_n} & \dots & \frac{\partial f}{\partial x_n \partial x_n} \end{bmatrix}_{n \times n}$$

HESSIAN MATRIX

(Symmetric)

$$(x-x_0)\,H(x_0)\,(x-x_0) = (x-x_0)^T \cdot H(x_0) \cdot (x-x_0)$$

$$[\,]_{1\times n} \cdot [\,]_{n\times n} \cdot [\,]_{n\times 1} = [\,]_{1\times n}\,[\,]_{n\times 1} = [\,]_{1\times 1}$$

$$a \cdot b = a_1 b_1 + \dots + a_n b_n$$

$$= a^T \cdot b$$

$$= \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}$$

$$[\,]_{1\times n} \cdot [\,]_{n\times 1}$$

(2) a) $f(x,y) = e^{x+y}$, $(0,0) = (x_0, y_0)$

$$\frac{\partial f}{\partial x} = e^{x+y} = \frac{\partial f}{\partial y}, \quad \nabla f(0,0) = (1,1)$$

$$\frac{\partial^2 f}{\partial x^2} = e^{x+y} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

$$H(0,0) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$f(x,y) \approx \underbrace{f(0,0)}_{=1} + \underbrace{\nabla f(0,0)\cdot(x,y)}_{x+y} + \frac{1}{2}\begin{bmatrix} x \\ y \end{bmatrix}^T \begin{bmatrix} x+y \\ x+y \end{bmatrix}$$

$$H(0,0)\cdot(x,y) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x+y \end{bmatrix}$$

$$f(x,y) \approx 1 + x + y + \frac{1}{2}[x(x+y) + y(x+y)]$$

$$= 1 + x + y + \frac{1}{2}(x+y)^2$$

→ 2nd order Taylor pol.

$$1 + (x+y) + \frac{1}{2}(x+y)^2$$

$$x+y = t, \quad e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots + \frac{t^k}{k!} + \dots$$

③ $D=\begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}_{n\times n}$, $f(x)=\frac{1}{2}x^T Dx = \frac{1}{2}(d_1x_1^2+d_2x_2^2+\dots+d_nx_n^2)$

$x\in\mathbb{R}^n$

$Dx=\begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}_{n\times n}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}=\begin{bmatrix} d_1x_1 \\ d_2x_2 \\ \vdots \\ d_nx_n \end{bmatrix}$

$x^TDx=\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}_{1\times n}\begin{bmatrix} d_1x_1 \\ d_2x_2 \\ \vdots \\ d_nx_n \end{bmatrix}_{n\times 1}=d_1x_1^2+d_2x_2^2+\dots+d_nx_n^2$

- $\nabla f(x)=\left(\frac{\partial f}{\partial x_1},\frac{\partial f}{\partial x_2},\dots,\frac{\partial f}{\partial x_n}\right)$

$=(d_1x_1, d_2x_2,\dots,d_nx_n)$

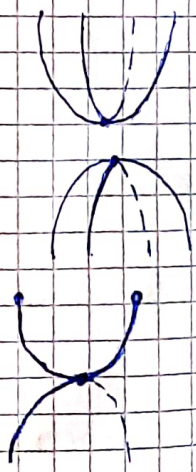
$=Dx$

$D(x+y)=Dx+Dy$

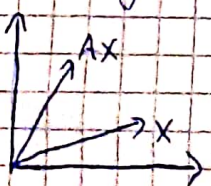
- $H(x)=\begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}=D$

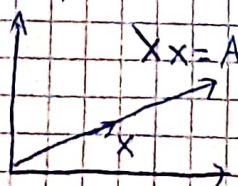
Critical point: $\nabla f(x_0)=0$

- $H(x_0)$ is positive definite: $x^TH(x_0)x>0$ $\forall x\in\mathbb{R}^n$
  $\Rightarrow x_0$ - MIN
- $H(x_0)$ is negative definite: $x^TH(x_0)x<0$ $\forall x\in\mathbb{R}^n$
  $\Rightarrow x_0$ - MAX
- $H(x_0)$ is indefinite $\Rightarrow x_0$ - SADDLE POINT



- Eigenvalues of $H(x_0)$ are positive $\Rightarrow H(x_0)$ is pos. def. $\Rightarrow$ MIN
  negative $\Rightarrow H(x_0)$ is neg. def. $\Rightarrow$ MAX
- Some eigenvalues >0, some are <0 $\Rightarrow H(x_0)$ is indefinite $\Rightarrow$ SADDLE POINT





Eigenvalue

$Ax=\lambda\cdot x$, λ - eigenvalue

x - eigenvector $x\neq 0$

$(A - \lambda I_n)x = 0$, $x \neq 0$

$\det(A - \lambda I_n) = 0$

$f(x, y) = x^2 - y^2$

Critical points: $\nabla f = 0$

$\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = -2y$

Critical point $(0,0)$, $\nabla f(0,0) = 0$

$\frac{\partial^2 f}{\partial x^2} = 2$, $\frac{\partial^2 f}{\partial y^2} = -2$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 0$

$H(x,y) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} = H(0,0)$

"diag matrices have λ values on the diagonal"

$\begin{vmatrix} 2-\lambda & 0 \\ 0 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) = 0$

$\lambda_1 = 2$, $\lambda_2 = -2 \Rightarrow H(0,0)$ is indefinite

$\Rightarrow (0,0)$ is a SADDLE POINT

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Lecture 12

Taylor's Theorem in 1d:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + \text{Remainder}$$

(small as $x-x_0$ is small)

Taylor's Theorem in nd:

Let $f: A \subseteq \mathbb{R}^n \to \mathbb{R}$ of class C^2, $x_0 \in A$, then

$$f(x) = f(x_0) + \nabla f(x_0)\cdot(x-x_0) + \frac{1}{2}(x-x_0)\cdot H(x_0)(x-x_0) + \text{Remainder}$$

Here $\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0)\right)$, $\nabla f(x_0)$ - gradient

$$H(x_0) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Hessian

This can also be written (taking $h = x - x_0$, $x = x_0 + h$) as:

$$f(x_0+h) = f(x_0) + \underbrace{\nabla f(x_0)\cdot h}_{\nabla f(x_0)^T \cdot h} + \frac{1}{2}\underbrace{h \cdot H(x_0)\cdot h}_{h^T \cdot H(x_0)\cdot h} + \text{Remainder}$$

$[\,]_{1\times n}\ [\,]_{n\times 1}$ $\quad [\,]_{1\times n}\ [\,]_{n\times n}\ [\,]_{n\times 1}$

Recall that $\underset{[\,]_{n\times 1}\ [\,]_{n\times 1}}{a \cdot b} = a_1 b_1 + \dots + a_n b_n = a^T b = [a_1\ a_2 \dots a_n]\begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}$

Proof: Let $g(t) = f(x_0 + t h)$. We have that $g(0) = f(x_0)$, $g(1) = f(x_0+h)$

- 1d Taylor expansion for g around 0:
  $$g(1) = g(0) + g'(0) + \frac{1}{2} g''(0) + \text{Remainder}$$
- Compute the derivatives of g using the Chain Rule:
  $$g'(t) = \frac{d}{dt} f(x_0 + t h) = \underset{\text{Chain Rule}}{\uparrow} \nabla f(x_0 + t h) \cdot \frac{d}{dt}(x_0 + t h) = \nabla f(x_0 + t h)\cdot h$$

$$\boxed{g'(0) = \nabla f(x_0)\cdot h}$$

- $$g''(t) = \frac{d}{dt} g'(t) = \frac{d}{dt}\left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(x_0 + t h)\cdot h_i\right)$$
  $$= \sum_{i=1}^n h_i \frac{d}{dt}\left(\frac{\partial f}{\partial x_i}(x_0 + t h)\right)$$

Chain Rule $\rightarrow$ $= \sum_{i=1}^n h_i \underbrace{\left[\nabla \frac{\partial f}{\partial x_i}(x_0 + t h)\cdot h\right]}_{v_i}$, where $v = \begin{bmatrix} \nabla \frac{\partial f}{\partial x_1}(\dots)\cdot h \\ \nabla \frac{\partial f}{\partial x_2}(\dots)\cdot h \\ \dots \\ \nabla \frac{\partial f}{\partial x_n}(\dots)\cdot h \end{bmatrix}$

$$= h \cdot v = h \cdot H(x_0 + t h)\, h = h^T \cdot H(x_0 + t h)\cdot h$$

$$\boxed{g''(0) = h \cdot H(x_0)\, h = h^T H(x_0)\cdot h}$$

- Recall that $g(1) = g(0) + g'(0) + \frac{1}{2} g''(0) + \text{Remainder}$

$$\boxed{f(x_0+h) = f(x_0) + \nabla f(x_0)\cdot h + \frac{1}{2} h^T H(x_0)\cdot h + \text{Remainder}}$$

Corollary: If x_0 is a critical point, $\nabla f(x_0)=0$, then:

- If $h^T H(x_0) h > 0$, $\forall h \in \mathbb{R}^n$, then $f(x_0+h) > f(x_0)$, x_0-MIN
  ↳ $H(x_0)$ is positive definite (eigenvalues >0)
- If $h^T H(x_0) h < 0$, $\forall h \in \mathbb{R}^n$, then $f(x_0+h) < f(x_0)$, x_0-MAX
  ↳ $H(x_0)$ is negative definite (eigenvalues <0)
- If $H(x_0)$ is indefinite, then x_0 - SADDLE POINT

$z = x^2 - y^2 = f(x,y)$

$(0,0)$ - Saddle point

$\nabla f(x,y) = (2x, -2y) = 0 \Rightarrow \begin{matrix} x=0 \\ y=0 \end{matrix}$

$$H(x,y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

eigenvalues are $\overset{>0}{2}; \overset{<0}{-2} \Rightarrow$ Saddle point

Applications: Linear regression

- Consider n data points (x_i, y_i), $i=\overline{1,n}$

We want to find a line $f(x) = \alpha x + \beta$ s.t.

(least squares error) $E = \sum_{i=1}^{n} |f(x_i) - y_i|^2 \to \min$

$$E(\alpha, \beta) = \sum_{i=1}^{n} (\alpha x_i + \beta - y_i)^2 \to \min$$

- Look for the critical points of $E(\alpha, \beta)$, i.e. $\frac{\partial E}{\partial \alpha} = 0 = \frac{\partial E}{\partial \beta}$

- $\frac{\partial E}{\partial \alpha} = \sum_{i=1}^{n} 2(\alpha x_i + \beta - y_i) x_i = 0$

$$\alpha \sum_{i=1}^{n} x_i^2 + \beta \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$$

- $\frac{\partial E}{\partial \beta} = \sum_{i=1}^{n} 2(\alpha x_i + \beta - y_i) = 0$

$$\alpha \sum_{i=1}^{n} x_i + \beta n = \sum_{i=1}^{n} y_i$$

- $\frac{\partial E}{\partial \alpha} = \frac{\partial E}{\partial \beta} = 0$ can be written as

$\Rightarrow \det > 0$ (Cauchy-Schwarz ineq.)

$$\begin{bmatrix} \sum_{i=1}^{n} x_i^2 & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & n \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} x_i y_i \\ \sum_{i=1}^{n} y_i \end{bmatrix}$$

• Applications - Gradient descent

- Start from an initial value x_0
- Want to minimize an objective function f

• Idea: We are at a point x_k

In which direction should we move from x_k?

Answer: the direction of fastest decrease (steepest decrease)

$$-\nabla f(x_k)$$

• Gradient descent:

$$\boxed{x_{k+1} = x_k - s\,\nabla f(x_k)}\,;\quad s = \text{step size (learning rate)},\ s>0,\ s = s(k)$$

• Exact line search: Choose the optimal step size s

Minimize: $\varphi(s) = f(x_{k+1}) = f(x_k - s\nabla f(x_k))$

$\varphi'(s) = 0 \Rightarrow$ optimal step size

• $\varphi'(s) = \frac{d}{ds} f(x_{k+1}) = \frac{d}{ds} f(x_k - s\nabla f(x_k))$

Chain Rule $\longrightarrow = \nabla f(x_k - s\nabla f(x_k)) \cdot \frac{d}{ds}(x_k - s\nabla f(x_k))$

$= \nabla f(x_{k+1}) \cdot (-\nabla f(x_k))$

• $\varphi'(s) = 0$ gives $\nabla f(x_{k+1}) \cdot \nabla f(x_k) = 0$

• Notice that $x_{k+1} = x_k - s\nabla f(x_k)$, $x_{k+1} - x_k = -s\nabla f(x_k)$

$x_{k+2} = x_{k+1} - s\nabla f(x_{k+1})$, $x_{k+2} - x_{k+1} = -s\nabla f(x_{k+1})$

• $\nabla f(x_{k+1}) \cdot \nabla f(x_k) = 0 \Rightarrow (x_{k+1} - x_k)(x_{k+2} - x_{k+1}) = 0$

$x_{k+1} - x_k$ perpendicular to $x_{k+2} - x_{k+1}$

Seminar 12

① $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$ $\quad X^T A X$ $\quad [\,]_{1\times n}[\,]_{n\times n}[\,]_{n\times 1}$ $\quad X^T = [x_1\ x_2 \cdots x_n]_{1\times n}$

(In Seminar 11 this question was for)

$A = \text{diag}.(d_1, \ldots, d_n) = \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$

Taylor Expansion in nD:

$$f(x_0+h) = f(x_0) + \underbrace{\nabla f(x_0)\cdot h}_{\text{linear term in } h} + \underbrace{\frac{1}{2} h^T H(x_0) h}_{\text{quadratic term in } h} + \text{Remainder}$$

Let $f(x) = \frac{1}{2} x^T A x$

$$f(x_0+h) = \frac{1}{2}(x_0+h)^T \cdot A \cdot (x_0+h) = \frac{1}{2} x_0^T A x_0 + \frac{1}{2} x_0^T A h + \frac{1}{2} h^T A x_0 + \frac{1}{2} h^T A h$$

$$f(x_0+h) \underset{\text{for } h}{=} f(x_0) + \underbrace{h^T A x_0}_{= h\,(Ax_0)} + \frac{1}{2} h^T A h \Rightarrow \nabla f(x_0) = Ax_0, \quad H(x_0) = A$$

Note that $\frac{1}{2} x_0^T A h \overset{\text{dot product}}{=} \frac{1}{2} x_0 \cdot (Ah) \overset{\text{commut.}}{=} \frac{1}{2} (Ah)\cdot x_0 \overset{\text{dot prod.}}{=} \frac{1}{2}(Ah)^T x_0$

Recall that $a \cdot b = a_1 b_1 + \ldots + a_n b_n = a^T b = [a_1\ a_2 \cdots a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} =$

$= b^T a = b \cdot a$

$(AB)^T = B^T A^T$

$$\frac{1}{2} x_0^T A h = \frac{1}{2}(Ah)^T x_0 = \frac{1}{2} h^T A^T x_0 \overset{A^T = A}{=} \frac{1}{2} h^T A x_0$$

Conclusion: For $f(x) = \frac{1}{2} x^T A x$, $\nabla f(x_0) = Ax_0$, $H(x_0) = A$

in 1d: $A \in \mathbb{R}$, $x \in \mathbb{R}$, $f(x) = \frac{A}{2} x^2$

$f'(x) = Ax$

$f''(x) = A$

Method 2 $f(x) = \frac{1}{2} x^T A x = \frac{1}{2} [x_1\ x_2 \ldots x_n] \begin{bmatrix} a_{11} & a_{12} \cdots a_{1n} \\ \cdots & \cdots \\ a_{n1} & a_{n2} \cdots a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

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Write $f(x)$ explicitly in term. of $x_1, \dots, x_n$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right)$$

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_2} \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

(2) Applications in Statistics, Optimisation, Machine Learning

Remark: If A is a square matrix, A invertible, then $\|Ax-b\|^2 \to \min$ is given by $Ax-b=0$ or $Ax=b$

$$x = A^{-1} \cdot b$$

Proof: Let $f: \mathbb{R}^n \to \mathbb{R}$, $f(x) = \|Ax-b\|^2$ (*)

We want to minimise $f \Rightarrow$ we look for $\nabla f(x) = 0$

- $f(x) = \|Ax-b\|^2 = (Ax-b)\cdot(Ax-b) = (Ax-b)^T(Ax-b)$

$(AB)^T = B^T A^T$

$$= (Ax)^T Ax - \underbrace{(Ax)^T b}_{Ax \cdot b} - \underbrace{b^T (Ax)}_{b \cdot Ax} + b^T b$$

$$= x^T A^T \cdot Ax - 2\, b^T \cdot (Ax) + b^T b$$

$b^T(Ax) = (b^T A)x = (A^T b)^T x$

$$= \underbrace{x^T (A^T A) x}_{\text{quadratic in } x} - \underbrace{2 (A^T b) \cdot x}_{\text{linear in } x} + b^T b = f(x)$$

- Looking for ($\nabla f(x) = 0$, that means $\nabla f(x) \to 0$)

$$\nabla f(x) = 2 A^T A x - 2 A^T b = 0$$

$$A^T A x - A^T b = 0, \quad A^T A x = A^T b \to \text{linear system}$$

$[\,]_{n \times m} [\,]_{m \times n} = [\,]_{n \times n}$ $\quad$ $[\,]_{n \times m} [\,]_{m \times 1} = [\,]_{n \times 1}$

- Take $H(x) = 2A^T A$ is positive definite

$$h^T A^T A h > 0, \quad (h^T A^T) A h = (Ah)^T Ah = (Ah)\cdot(Ah)$$

$$= \|Ah\|^2 > 0$$

- $H(x)$ positive definite $\Rightarrow$ Min point

• Conclusion $\min_{x \in \mathbb{R}^n} \|Ax-b\|^2$ is given by $A^TAx = A^Tb$

If A^TA is invertible, then there is a unique sol. x

Not for Exam → If A^TA is not invertible, then we have to add regularization

$$\min_{x \in \mathbb{R}} \left(\|Ax-b\|^2 + \alpha \|x\|^2 \right)$$

③ $g(x) = c$ → Level Set

Lagrange function: $L(x, \lambda) = f(x) - \lambda (g(x) - c)$ ← Lagrange multiplier

constraint $g(x) = c$ is satisfied

$$\lambda (g(x) - c) = 0$$

$$L(x, \lambda) = f(x)$$

$$\nabla_x L = 0 \quad , \quad \frac{\partial L}{\partial x} = 0$$

a) $f(x,y) = x^2 + y^2$

$g(x,y) = x - y$

constraint: $g(x,y) + 1 = 0$

Minimize / Maximize $f(x,y)$ subject to $g(x,y) + 1 = 0$

Define the Lagrange function $L(x,y,\lambda)$

$$L(x,y,\lambda) = f(x,y) - \lambda (g(x,y) + 1)$$

↳ constraint

~~• Critical points of L:~~

$$L(x,y,\lambda) = x^2 + y^2 - \lambda (x - y + 1)$$

• Critical points of L:

$$0 = \frac{\partial L}{\partial x} = 2x - \lambda \Rightarrow 2x = \lambda \Rightarrow x = \frac{\lambda}{2} \Rightarrow \boxed{x = -\frac{1}{2}}$$

$$0 = \frac{\partial L}{\partial y} = 2y + \lambda \Rightarrow 2y = -\lambda = -2x \Rightarrow y = -x \Rightarrow y = -\frac{\lambda}{2} \Rightarrow \boxed{y = \frac{1}{2}}$$

$$0 = \frac{\partial L}{\partial \lambda} = -(x - y + 1) \Rightarrow x - y + 1 = 0 \Rightarrow \lambda + 1 = 0 \Rightarrow \boxed{\lambda = -1}$$

• There is a unique critical point $(-\frac{1}{2}, \frac{1}{2}, -1)$

b) Min/Max $(x+y)^2$ subject to $x^2+y^2=1$

Lagrange function $L(x,y,\lambda) = (x+y)^2 - \lambda(x^2+y^2-1)$

Critical point of L:

$$0 = \frac{\partial L}{\partial x} = 2(x+y) - 2\lambda x \Rightarrow x+y = \lambda x \quad (1)$$

$$0 = \frac{\partial L}{\partial y} = 2(x+y) - 2\lambda y \Rightarrow x+y = \lambda y \quad (2)$$

$$\Rightarrow \boxed{x = y}$$

$$0 = \frac{\partial L}{\partial \lambda} = x^2+y^2-1 \Rightarrow x^2+y^2 = 1 \Rightarrow 2x^2 = 1, \; x^2 = \frac{1}{2}$$

$$\boxed{x = \frac{\sqrt{2}}{2} = y} \qquad \boxed{x = -\frac{\sqrt{2}}{2} = y}$$

$$2x = \lambda x \Rightarrow \boxed{\lambda = 2}$$

Graph: $z = f(x,y) = (x+y)^2$

Constraint: $x^2+y^2 = 1$

$(1)+(2) \Rightarrow 2(x+y) = \lambda(x+y)$

I. $\lambda = 2 \Rightarrow x = y \Rightarrow x^2+y^2 = 1 \Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2}$

$\boxed{x = \frac{\sqrt{2}}{2} = y}$ $\qquad$ $\boxed{x = -\frac{\sqrt{2}}{2} = y}$

II. $x+y = 0 \Rightarrow x = -y$

$x^2+y^2 = 1 \Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2}$

$\boxed{x = \frac{\sqrt{2}}{2}, \; y = -\frac{\sqrt{2}}{2}}$ $\qquad$ $\boxed{x = -\frac{\sqrt{2}}{2}, \; y = \frac{\sqrt{2}}{2}}$

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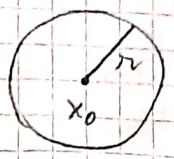
Lecture 9 (Recuperare)

$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$

$B(x^0, r) := \{x \in \mathbb{R}^n / \|x - x^0\| < r\}$
↳ open ball of center x^0 radius r

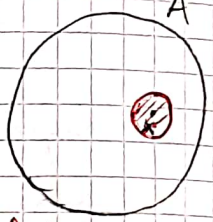


$d(x, x^0) = \|x - x^0\|$ ↳ distance

$\overline{B}(x^0, r) := \{x \in \mathbb{R}^n / \|x - x^0\| \leq r\}$
↳ closed ball of center x^0 radius r

• Recall that in $\mathbb{R}$, a set A is open if $\forall x \in A, \exists (x-r, x+r)$ s.t. $(x-r, x+r) \subseteq A$

Def: Let $A \subseteq \mathbb{R}^n$. We say that A is open if
$\forall x \in A, \exists\, B(x, r) \subseteq A$



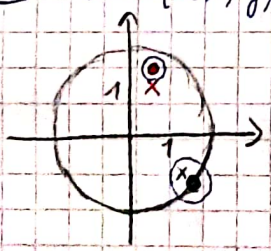
Def: Let $A \subseteq \mathbb{R}^n$. We define
$\text{int}(A) := \{x \in \mathbb{R}^n / \exists\, B(x, r) \subseteq A\}$ → interior of A

Def: $\text{cl}(A) := \{x \in \mathbb{R}^n / \forall B(x, r),\ B(x, r) \cap A \neq \emptyset\}$
↳ closure of A

Def: A set is closed if $A = \text{cl}(A)$

Def: $\text{bd}(A) := \{x \in \mathbb{R}^n / \forall B(x, r),\ B(x, r) \cap A \neq \emptyset$ and $B(x, r) \cap {}^cA \neq \emptyset\}$

ex: $A = \{(x, y) \in \mathbb{R}^2 / x^2 + y^2 < 1\}$

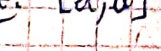


$\text{int}(A) = A$
$\text{bd}(A) = C(0,1) = \{(x, y) / x^2 + y^2 = 1\}$
$\text{cl}(A) = A \cup \text{bd}(A) = \{(x, y) / x^2 + y^2 \leq 1\}$

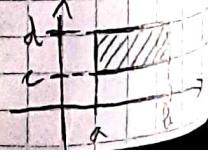
Def: Let $A \subseteq \mathbb{R}^n$. We say that A is bounded if $\exists\, r > 0$ s.t. $\|x\| < r, \forall x \in A$. $(A \subseteq B(r))$

Def: $A \subseteq \mathbb{R}^n$ is compact if A is bounded and closed

ex: $[a, b]$ is compact, $[a, b] \times [c, d]$ is compact

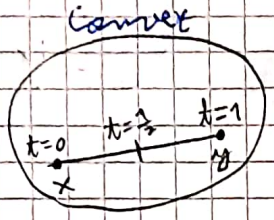


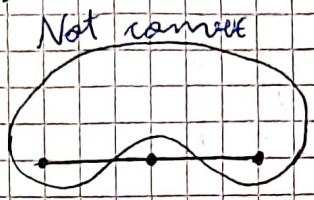
$\{(x, y) / a \leq x \leq b,\ c \leq y \leq d\}$



Def: $A \subseteq \mathbb{R}^n$. We say that A is convex if $\forall x, y \in A$, the segment joining them is inside A

$$(1-t)x + ty, \quad t \in [0,1]$$





Sequences in $\mathbb{R}^n$

Let $(x^k)_{k \in \mathbb{N}}$ is a sequence in $\mathbb{R}^n$ $(x^1, x^2, \dots, x^k, \dots \in \mathbb{R}^n)$

k-index for the sequence $\quad x^k = (x_1^k, x_2^k, \dots, x_n^k) \in \mathbb{R}^n$

Def: (x^k) converges to $x \in \mathbb{R}^n$ if $\lim\limits_{k \to \infty} \|x^k - x\| = 0$

ex: $x^k = \left(\frac{1}{k}, \frac{1}{k^2}\right)$, $\lim\limits_{k \to \infty} x^k = (0,0)$.

Thm: (x^k) converges to $x \in \mathbb{R}^n \iff x_i^k \to x_i$, $\forall i = \overline{1,n}$

Proof: „$\Rightarrow$" Let $i \in \{1, \dots, n\}$, $|x_i^k - x_i| = \sqrt{(x_i^k - x_i)^2} \leq$

$$\leq \sqrt{(x_1^k - x_1)^2 + \dots + (x_i^k - x_i)^2 + \dots} = \|x^k - x\|$$

Hence $|x_i^k - x_i| \leq \|x^k - x\|$, where $|x_i^k - x_i| \to 0$ and $\|x^k - x\| \to 0$

„$\Leftarrow$" For any $i \in \{1, \dots, n\}$, $|x_i^k - x_i| \to 0$

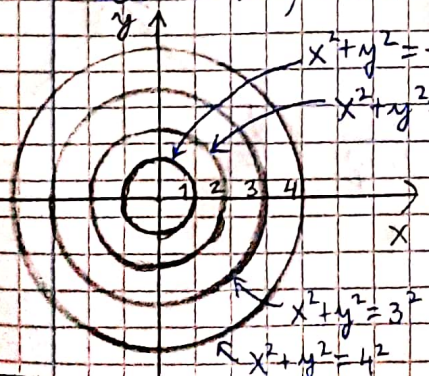
$\|x^k - x\| = \sqrt{(x_1^k - x_1)^2 + \dots + (x_n^k - x_n)^2} \to 0$, where $(x_1^k - x_1)^2 \to 0$ and $(x_n^k - x_n)^2 \to 0$, hence $x^k \to x$

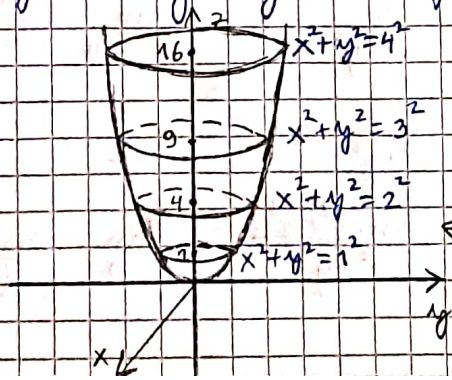
Functions of several variables

Let $A \subseteq \mathbb{R}^n$, let $f: A \to \mathbb{R}$ (Scalar function of several variables)

ex: $f: \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x^2 + y^2$

Let $c \geq 0$, then $\{(x,y) \in \mathbb{R}^2 / f(x,y) = x^2 + y^2 = c\} \to$ a circle





Graph of f is (x,y,z)

$z = f(x,y)$

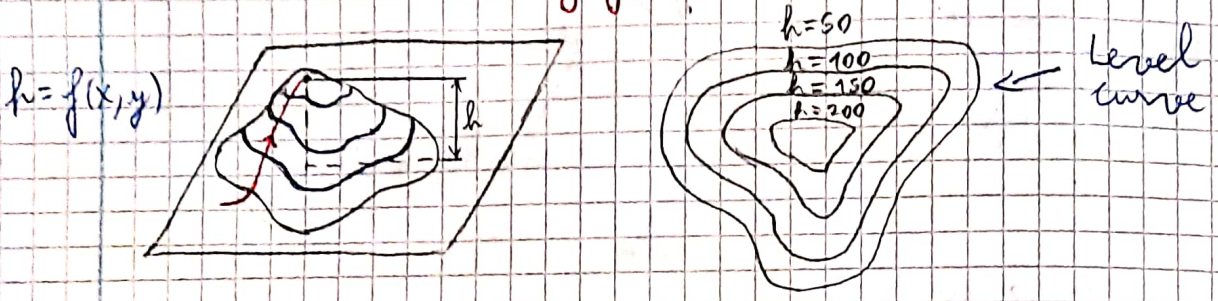
$z = c$ (constant)

Paraboloid

Def: Let $c \in \mathbb{R}$ constant, let $A \subseteq \mathbb{R}^n$, $f: A \to \mathbb{R}$

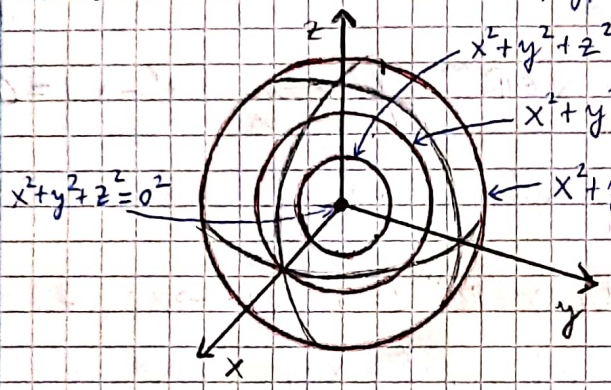
$L_c = \{x \in A \mid f(x) = c\}$

↳ Level set of f



ex: $f: \mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = x^2+y^2+z^2$

Level sets are $L_c = \{(x,y,z) \mid x^2+y^2+z^2 = c\}$ → spheres (Level surface)



Limit of a function

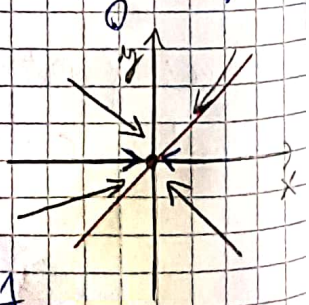
Let $A \subseteq \mathbb{R}^n$, $f: A \to \mathbb{R}$, $x^0 \in A'$

If for any sequence (x^k), $x^k \to x^0$, we have that $\lim_{k\to\infty} f(x^k) = l \in \mathbb{R}$, then $\lim_{x\to x^0} f(x) = l$

ex: a) $\lim_{(x,y)\to(0,0)} \frac{x^2}{\sqrt{x^2+y^2}} = 0$

$\frac{x^2}{\sqrt{x^2+y^2}} \le \frac{x^2+y^2}{\sqrt{x^2+y^2}} = \sqrt{x^2+y^2}$, so $\frac{x^2}{\sqrt{x^2+y^2}} \le \sqrt{x^2+y^2}$ ↓ as $(x,y)\to(0,0)$

b) $\lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+y^2}$ doesn't exist!



Take $y=0$, $\frac{x^2}{x^2+y^2} = 1$, $\lim_{(x,0)\to(0,0)} \frac{x^2}{x^2+y^2} = 1$

$y=x$, $\frac{x^2}{x^2+y^2} = \frac{x^2}{2x^2} = \frac{1}{2}$, $\lim_{(x,x)\to(0,0)} \frac{x^2}{x^2+y^2} = \frac{1}{2}$

c) $\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} = \lim_{t\to 0} \frac{\sin t}{t} = 1$

Continuity

Let $A \subseteq \mathbb{R}^n$, $f: A \to \mathbb{R}$, $x^0 \in A \cap A'$

f is continuous at x_0 if $\lim_{x \to x^0} f(x) = f(x^0)$

Thm (Weierstrass)

If $A \subseteq \mathbb{R}^n$ compact set, $f: A \to \mathbb{R}$ is cont. then f is bounded and it attains its bounds

$(\exists \min f(A),\ \exists \max f(A))$

Partial derivatives

$f(x_1, x_2, \dots, x_n)$

Idea: Differentiate f with respect to one variable, while the other variables are kept fixed

$\frac{\partial f}{\partial x_i}$ - partial derivative w.r.t. the variable x_i

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \to 0} \frac{f(x_1, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_n)}{h}$$

$$= \lim_{h \to 0} \frac{f(x + h e_i) - f(x)}{h}, \quad e_i = (0, \dots, 0, \underset{\downarrow i}{1}, 0, \dots, 0)$$

ex: a) $f(x,y) = x^2 y + y^2$

$$\frac{\partial f}{\partial x}(x,y) = 2xy + 0, \quad \frac{\partial f}{\partial y}(x,y) = x^2 + 2y$$

Def: Gradient of a function:

$$\nabla f(x) := \left(\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right)$$